LECTURE 3 ANATOMY

Anatomy of Cardiovascular System (CVS)-Lymphatic System & Respiratory system

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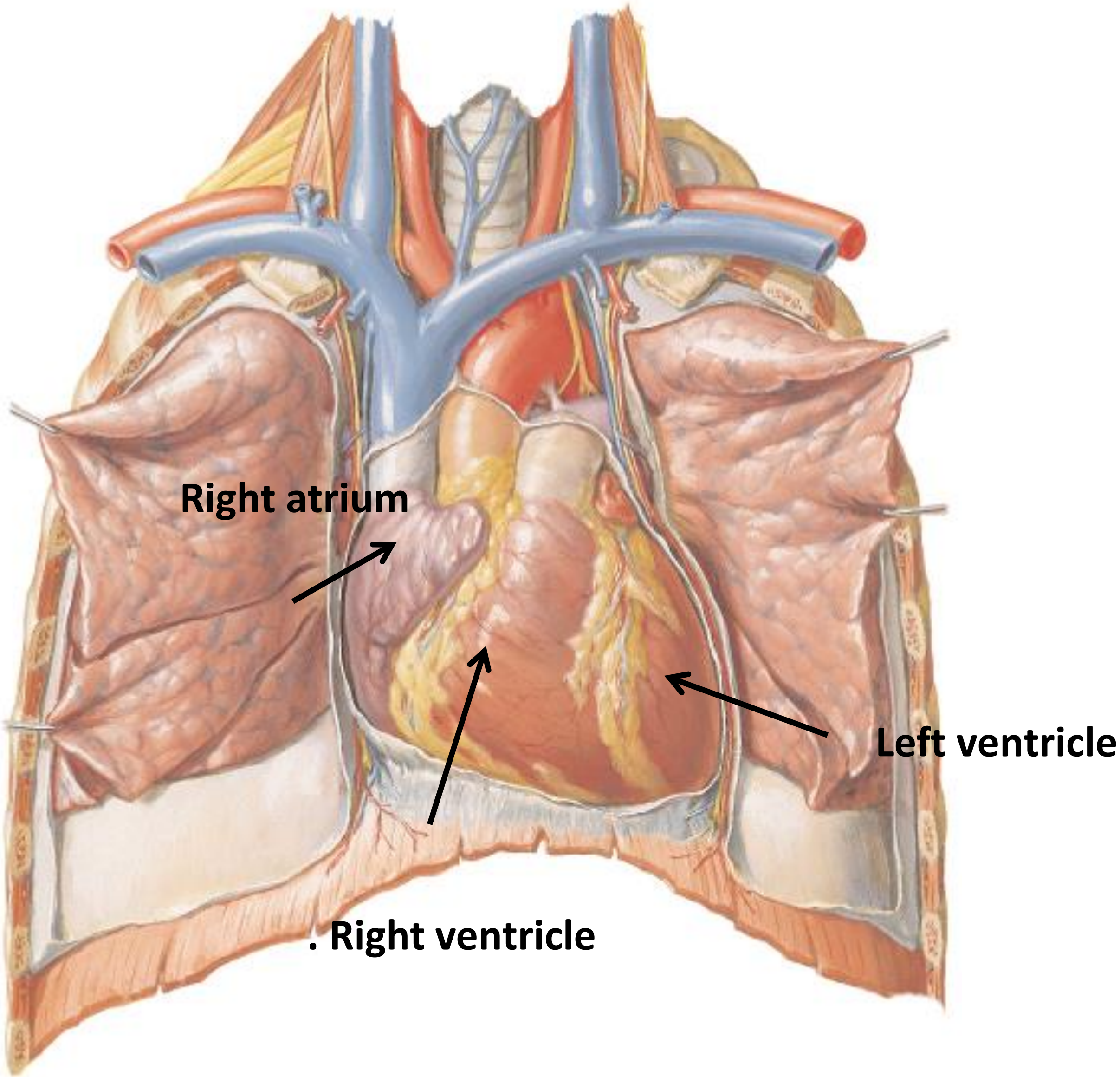
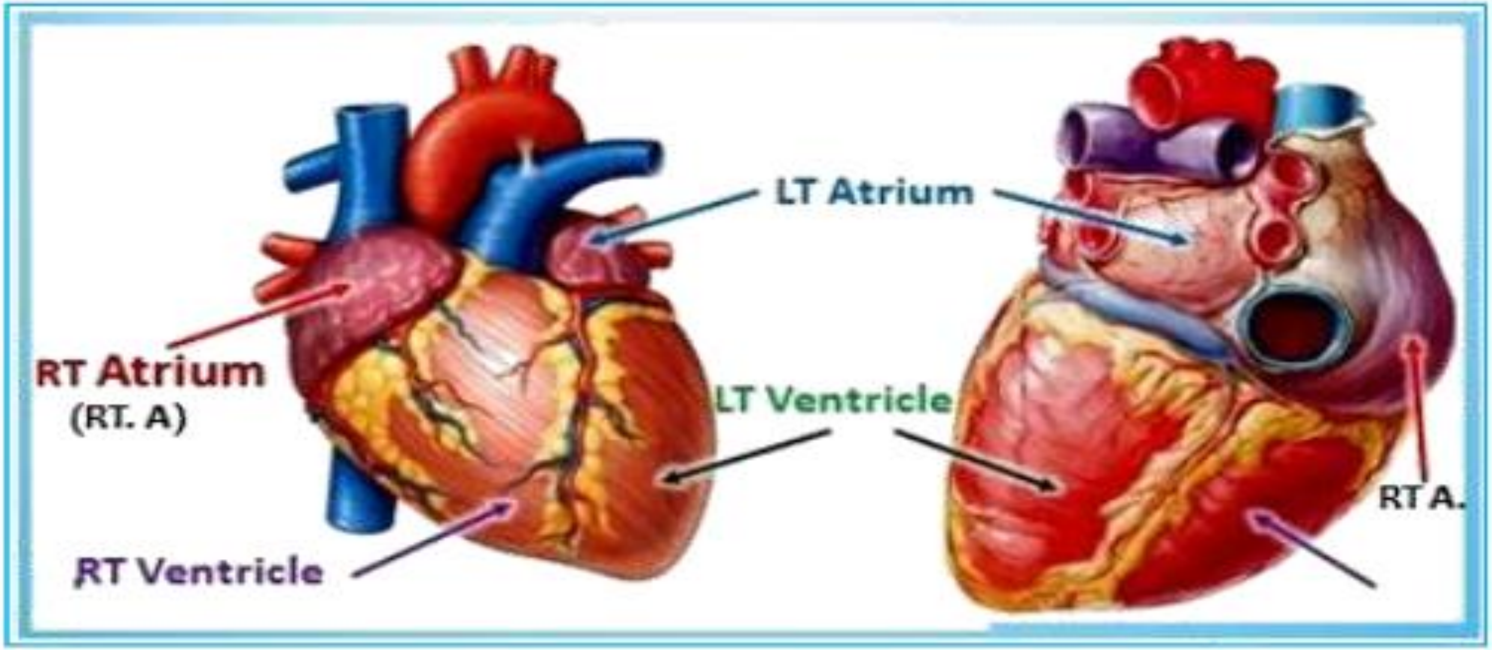
Cardiovascular System

- Cardiovascular system consists of heart and big vessels
- <u>**Heart consists of 4 chambers**</u>
- **2** atria and 2 ventricles

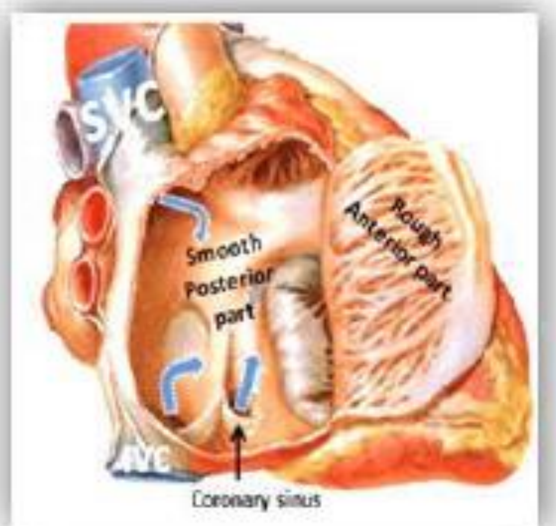
1. **Right atrium** receives venous blood
2. **Right ventricle** pumps blood to lungs
3. **Left atrium** receives oxygenated blood from lungs
4. **Left ventricle** pumps oxygenated blood to the whole body

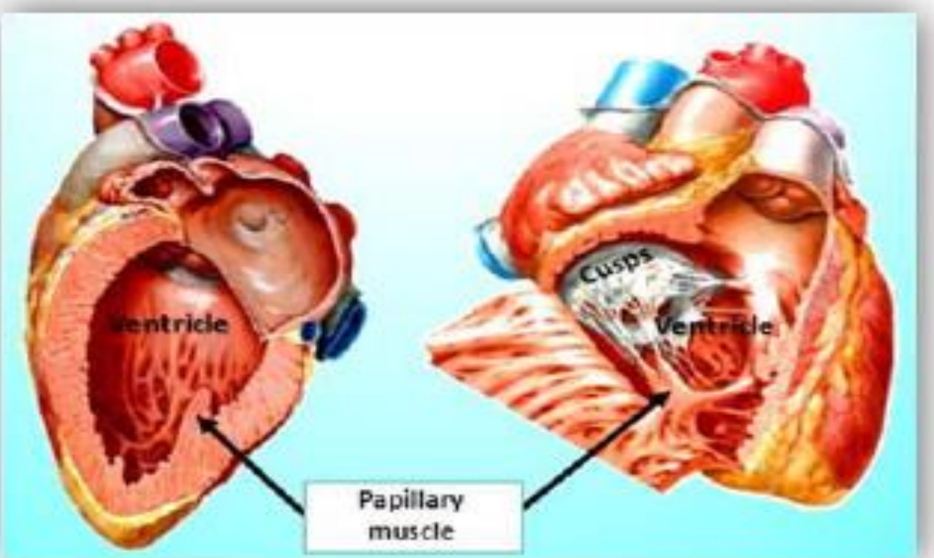
Heart



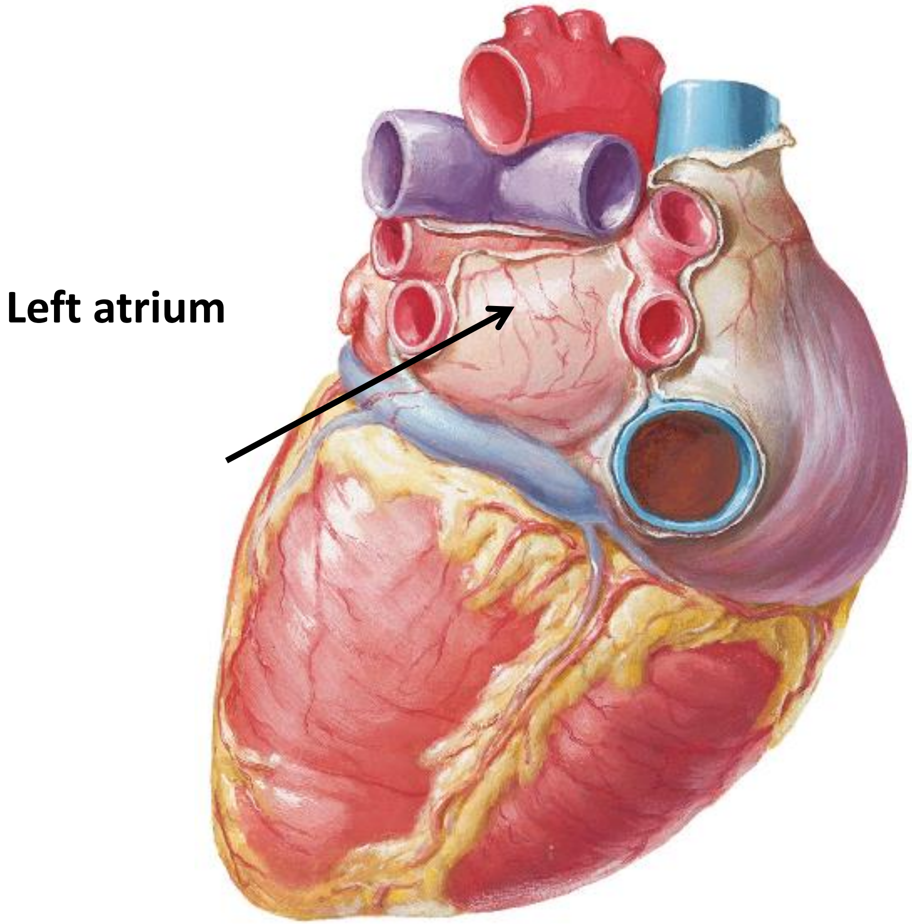


(Fig.13): Chambers of the heart





Heart



4 Big vessels attached to heart

- Superior vena cava
- Inferior vena cava
- Aorta
- Pulmonary trunk

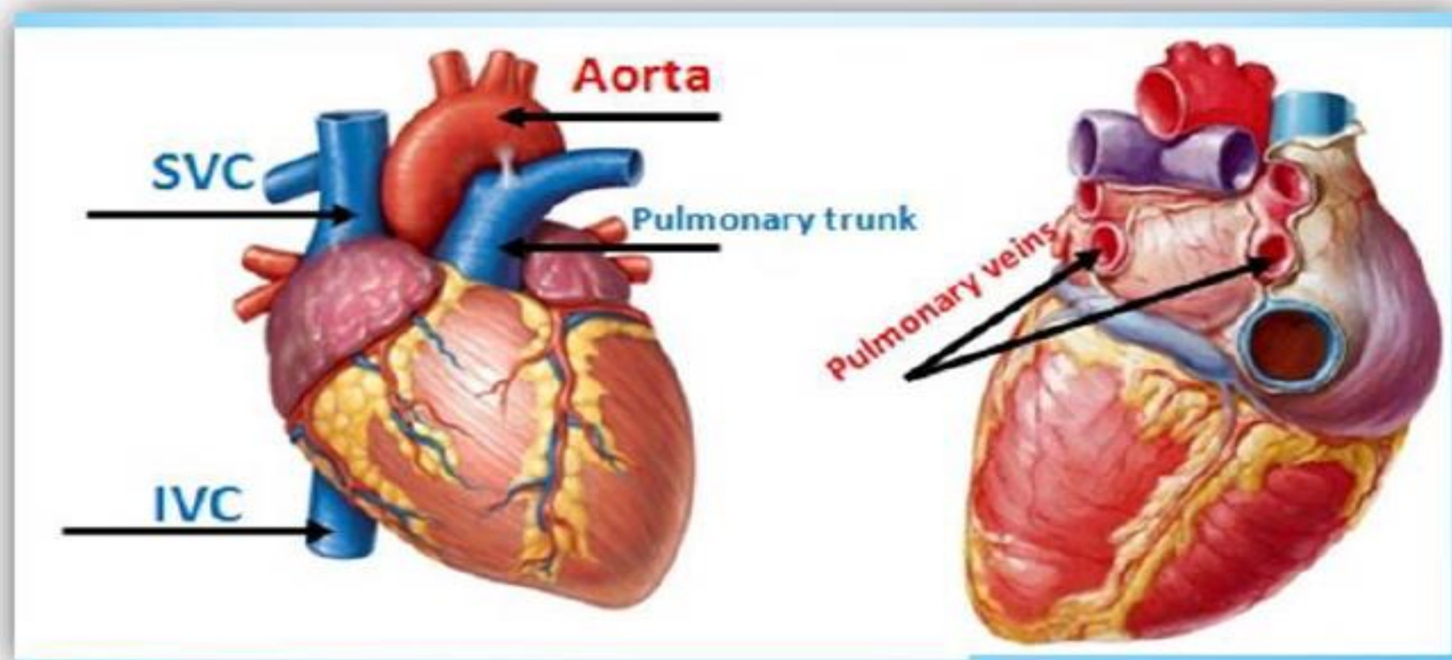
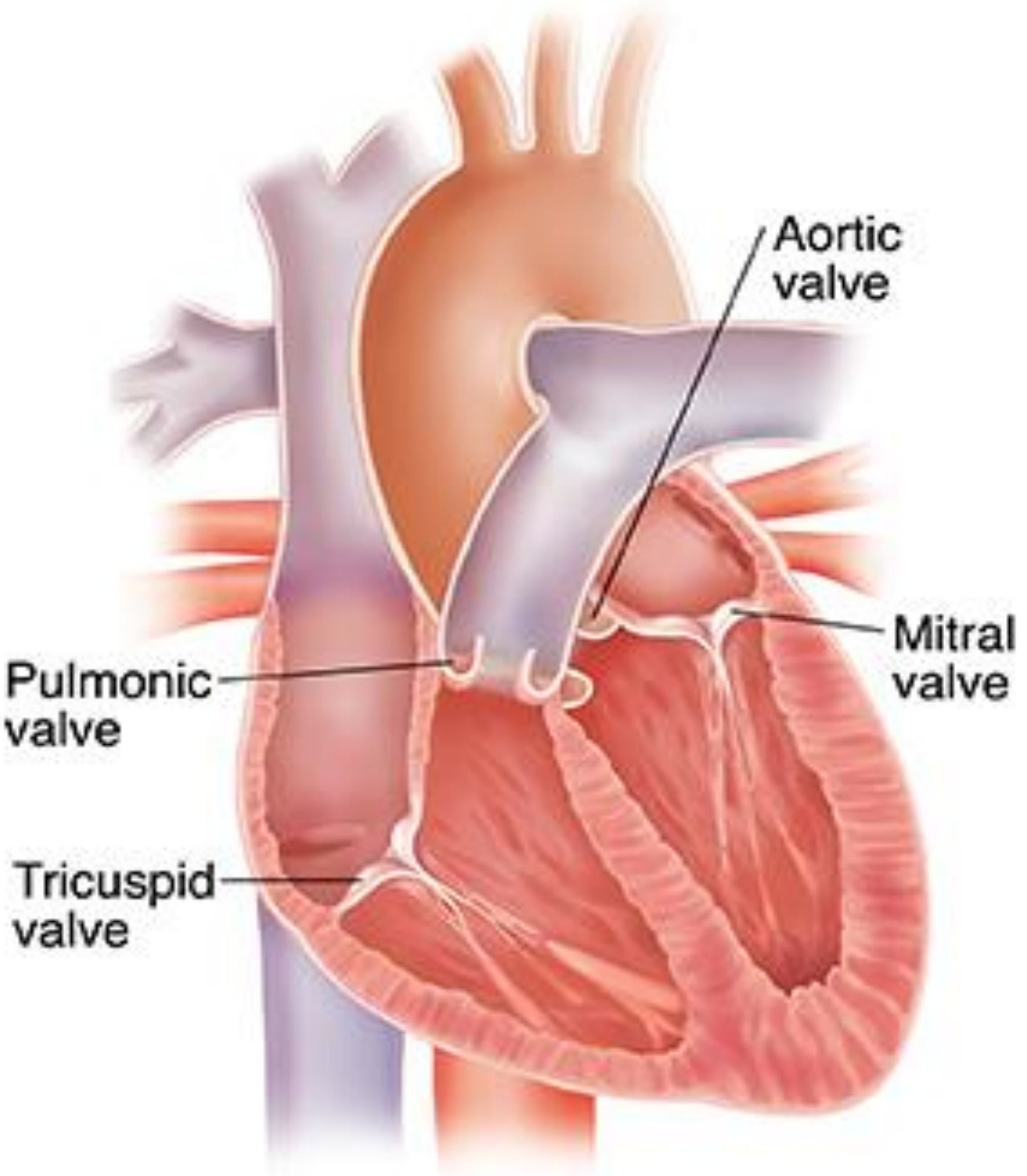


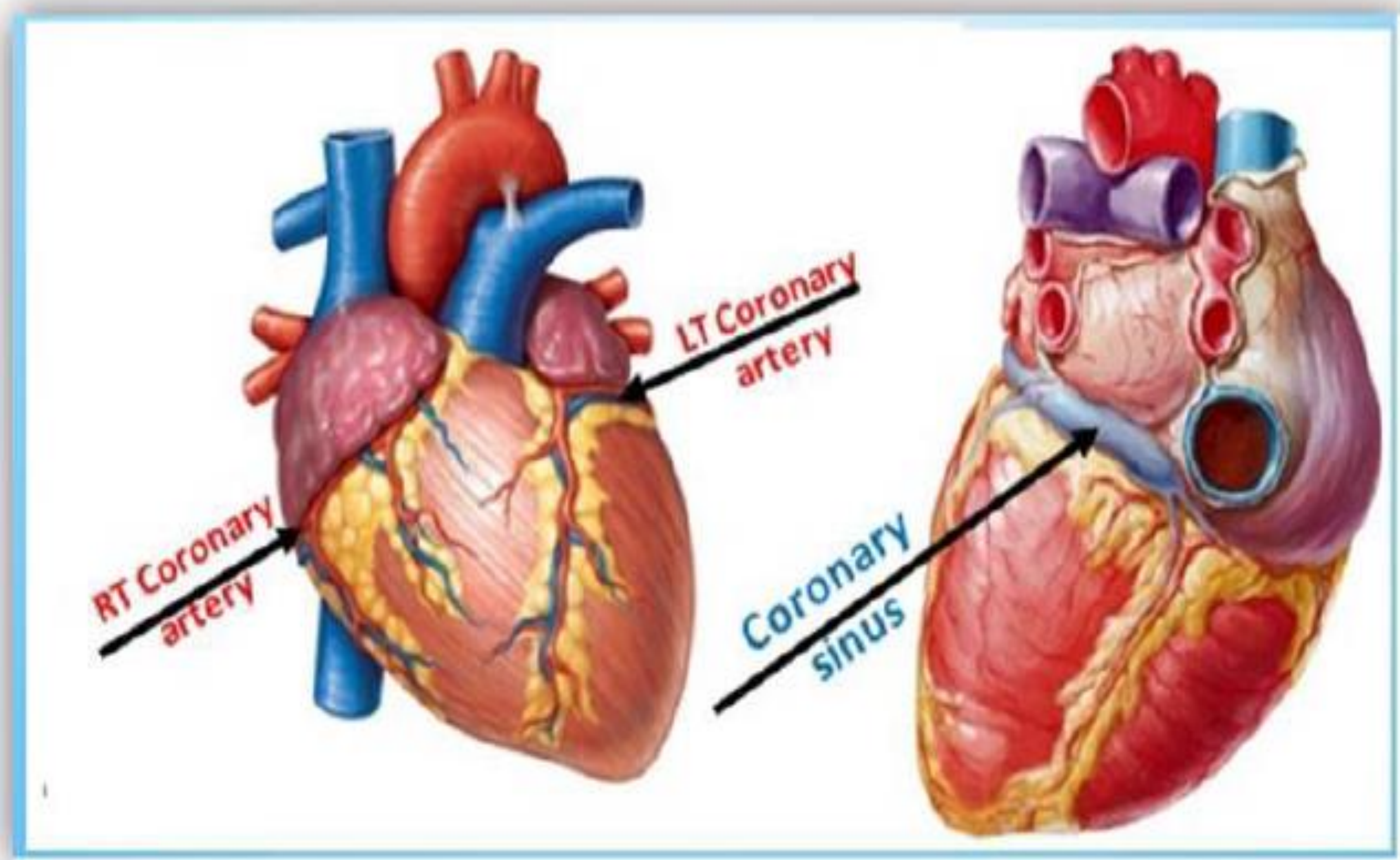
Fig.15): Vessels enter into & exit from the Heart

4 Valves

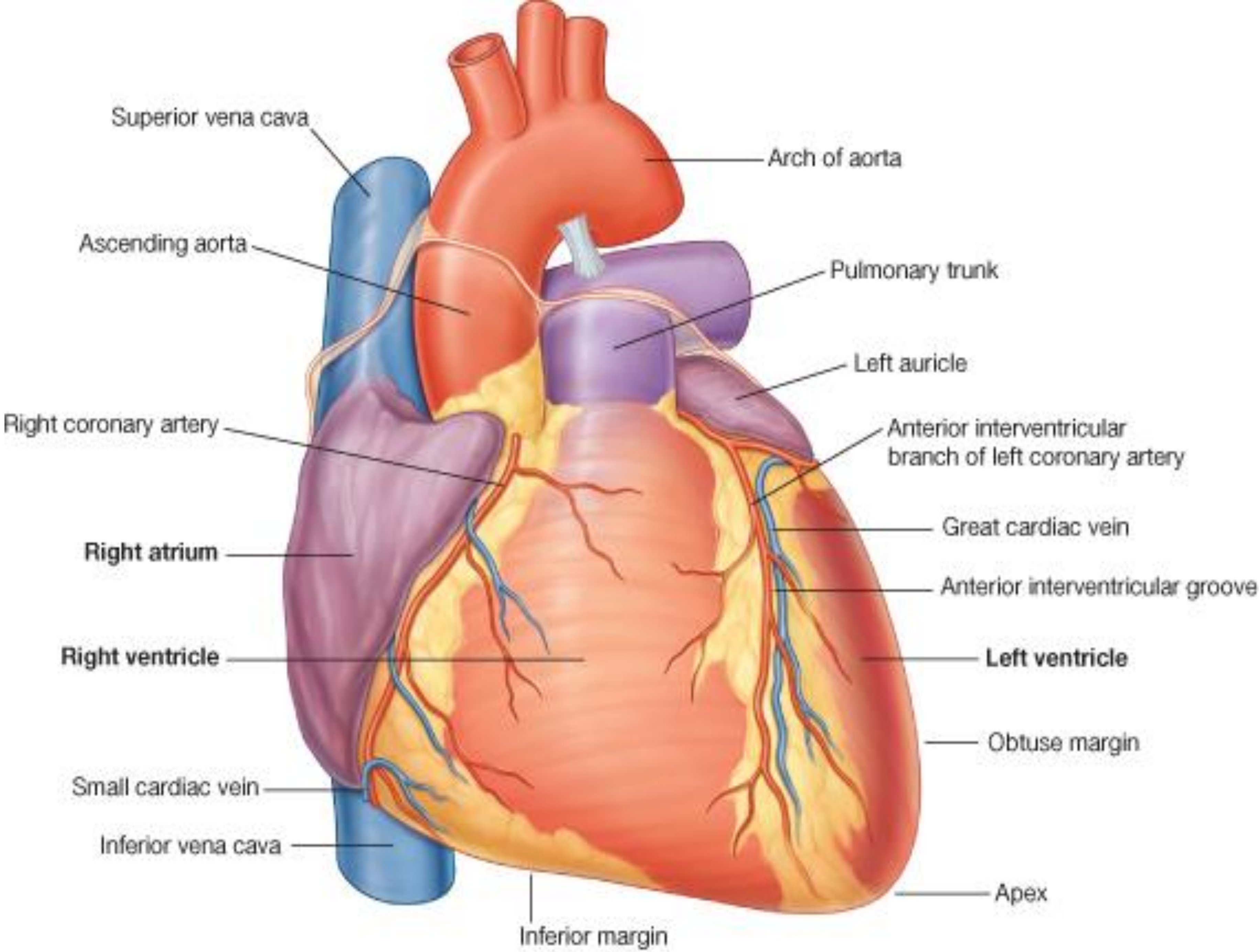
- **4 Valves =**

1. **Mitral**
2. **Tricuspid**
3. **Aortic**
4. **Pulmonary**





g.16): Coronary Arteries and Sinus of the Heart


Superior vena cava
Arch of aorta
Ascending aorta
Pulmonary trunk
Left auricle
Right coronary artery
Anterior interventricular
branch of left coronary artery
Great cardiac vein
Right atrium
Anterior interventricular groove
Right ventricle
Left ventricle
Obtuse margin
Small cardiac vein
Inferior vena cava
Apex
Inferior margin

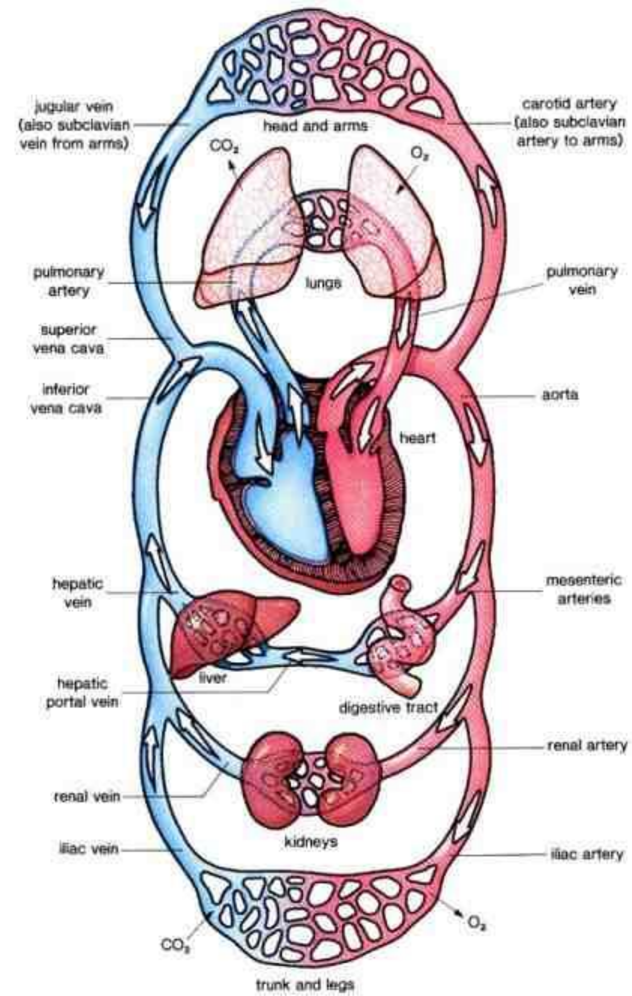
Types of Circulation

- **Systemic circulation**

Blood passes from left ventricle to aorta .Blood returns to right atrium via SVC, IVC

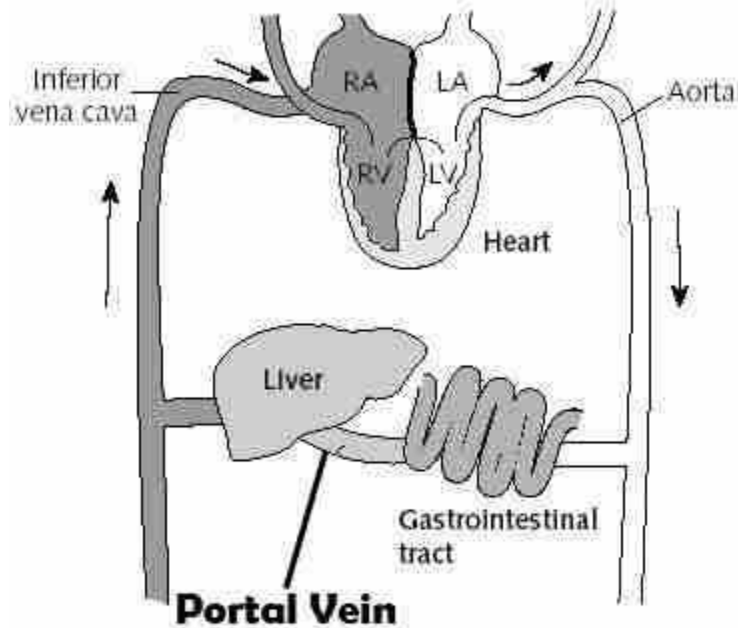
- **Pulmonary circulation**:

Blood passes from right ventricle to lungs via pulmonary artery then return to left atrium by 4 pulmonary veins



Portal Circulation

- **Venous blood** : **containing products of digestion & absorption** pass to liver via portal vein where liver **metabolizes** these products and blood **returns to systemic** circulation via **hepatic veins to IVC**



Blood Vessels

Arteries:

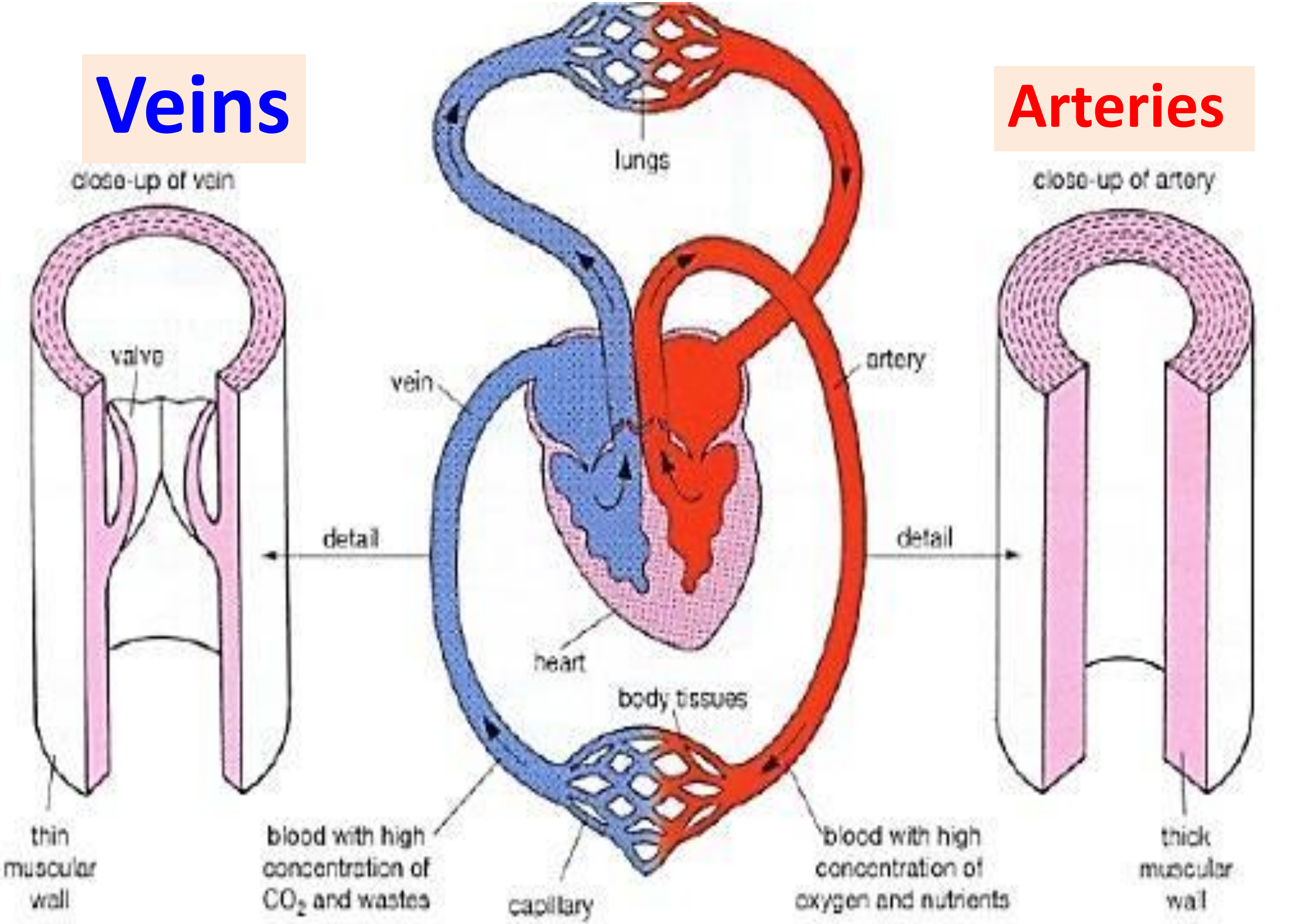
- Carry blood away from heart
- Thick wall contain muscles& elastic fibers
- Lumen: narrow
- Gives branches
- Carry oxygenated blood except pulmonary artery
- No valves

Veins

- Carry blood to heart
- Thin wall low content of muscles & elastic fibers
- Wide lumen
- Have tributaries
- Carry non oxygenated blood except pulmonary veins
- Have valves especially veins o lower limb to help in venous return

Veins

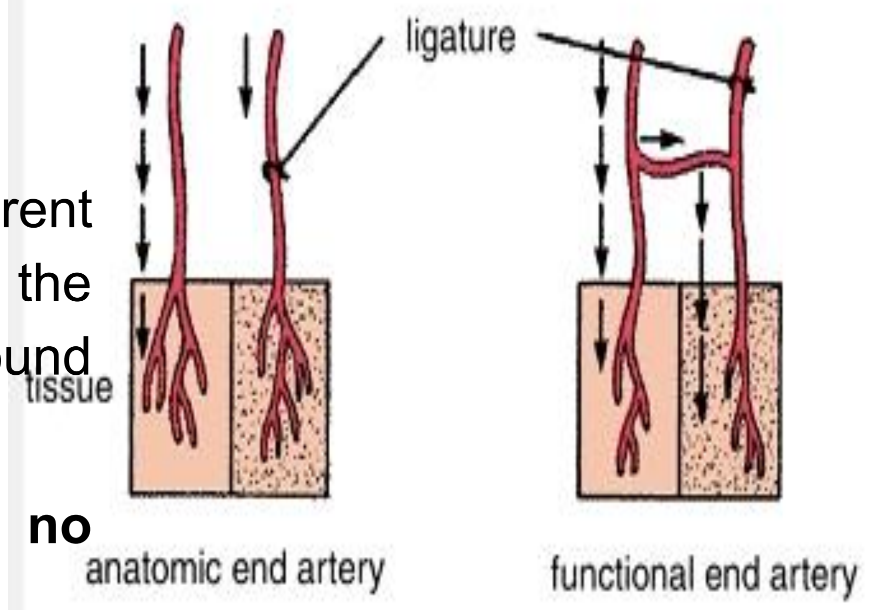
Arteries



1. Arteries

Arterial anastomosis :

Anastomosis is a connection between different arteries to maintain the arterial supply if the main artery is occluded. It is commonly found around the joints



End arteries: arteries that have **no anastomosis; o**r insufficient anastomosis..

End arteries are two types:

1) ***Anatomical end arteries***:

No connection exists at all between the arteries. If this artery is obstructed, damage of the area e.g. central retinal artery

2) *Functional end arteries*:

Anastomosis is present but it is not sufficient to compensate for occlusion of the main artery, e.g. coronary arteries**.**

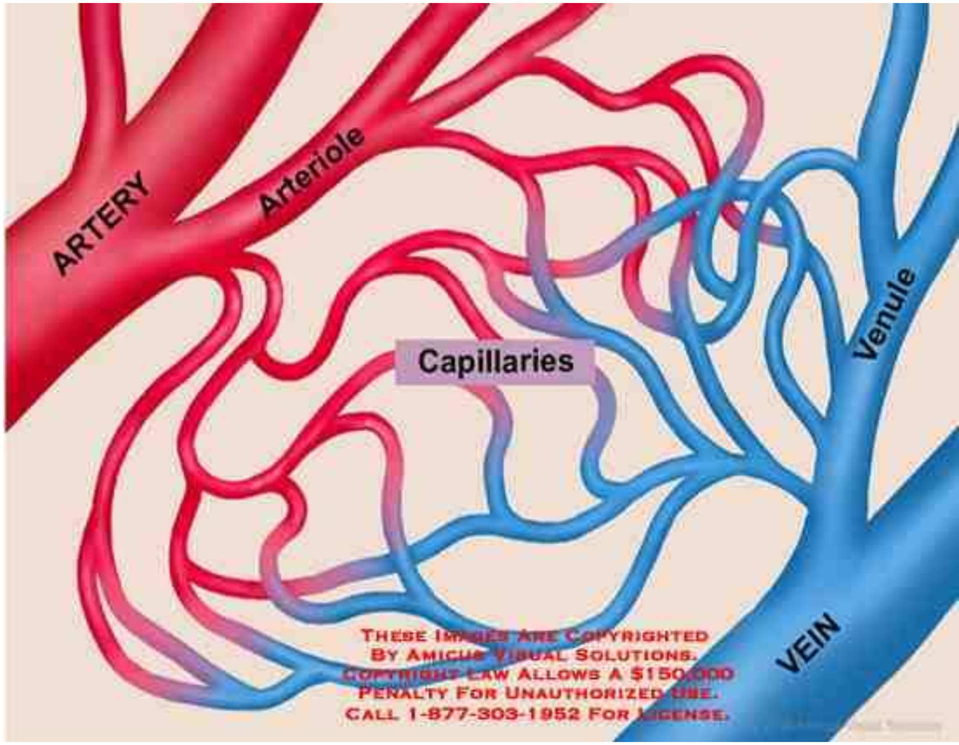
Anastomosis between arteries & veins

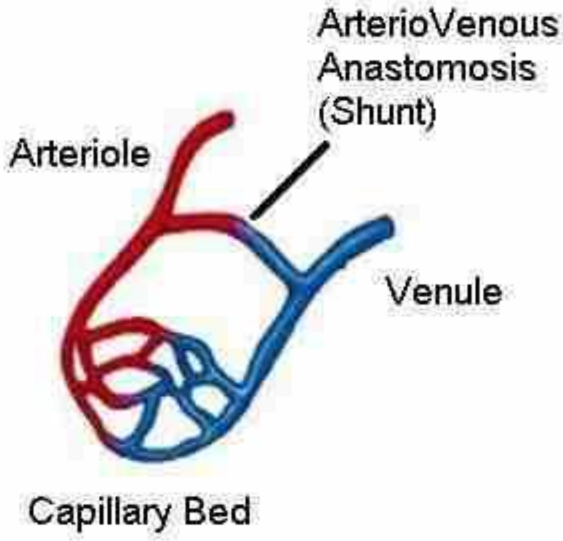
- **Capillaries**:

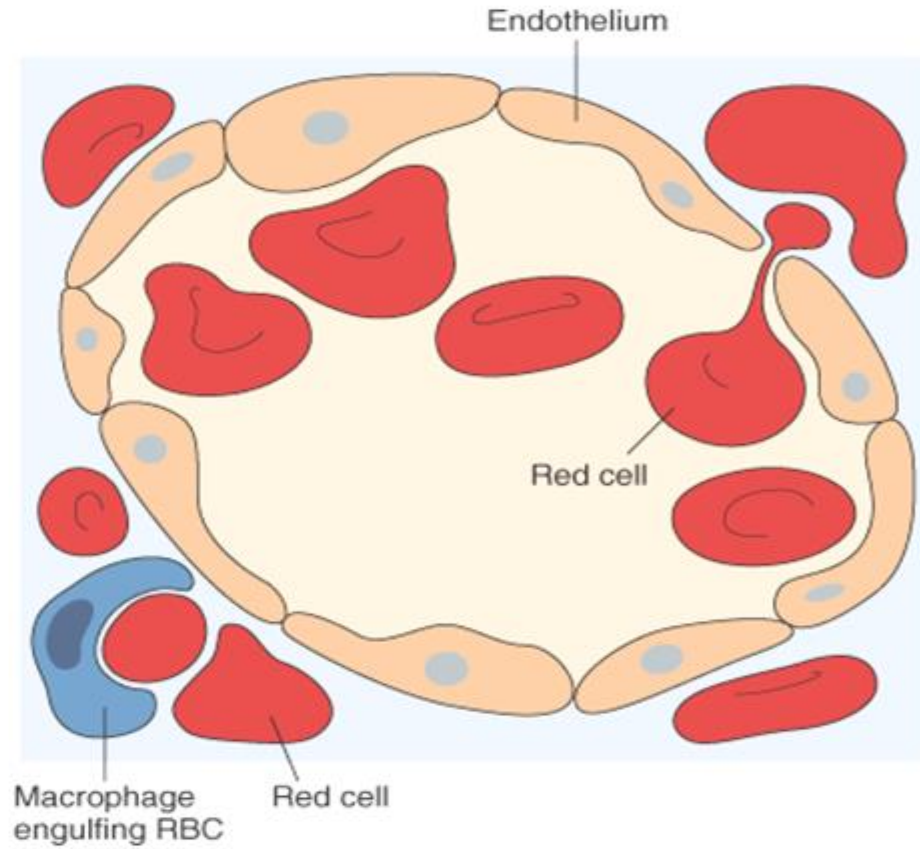
present all over body
connect small arterioles & small venules

- **Direct arteriovenous shunts:**

=connect arteries & veins
without capillaries **present in gut, skin , fingers, toes , nose, lips & ears**





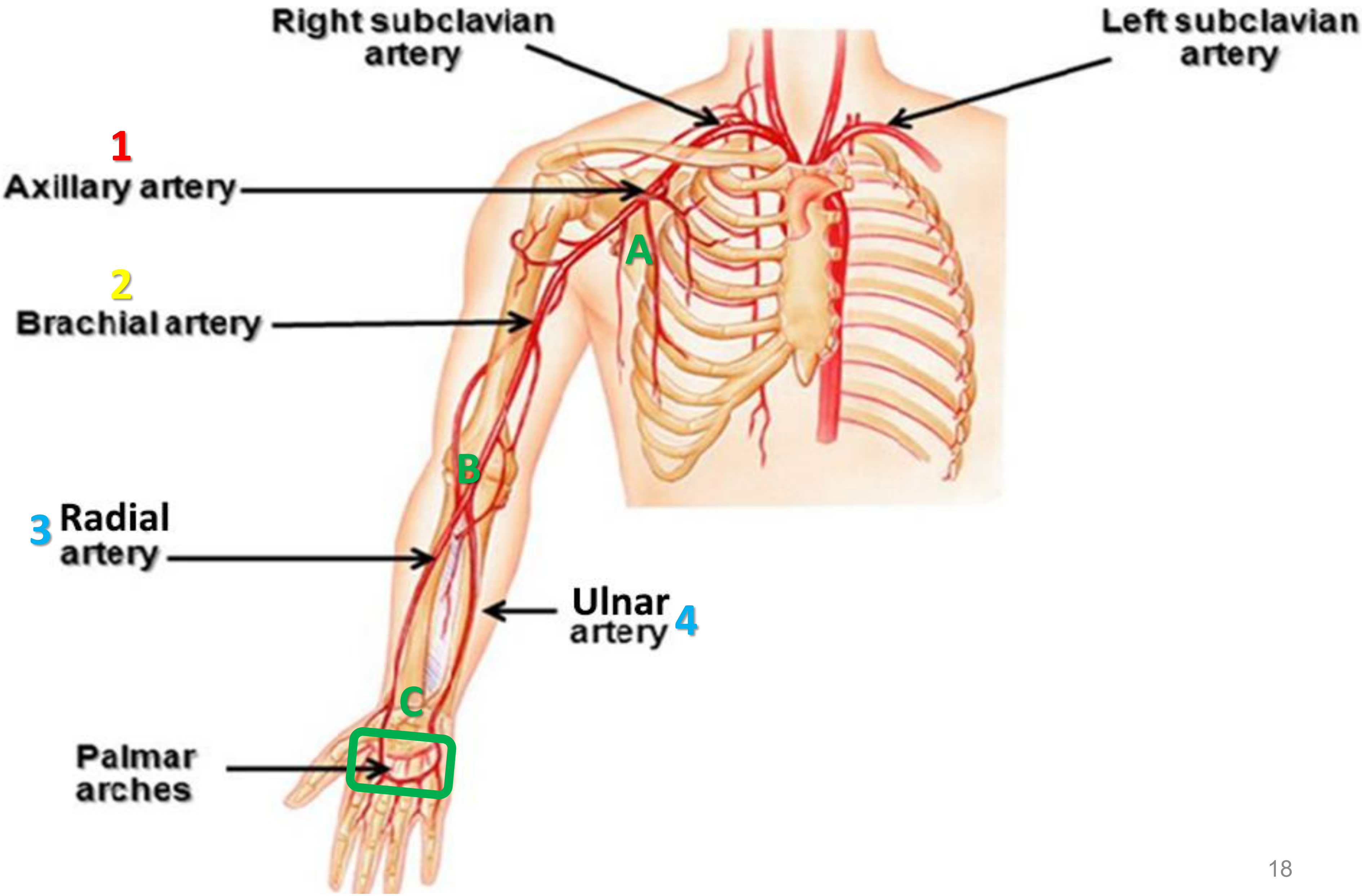


Sinusoids :

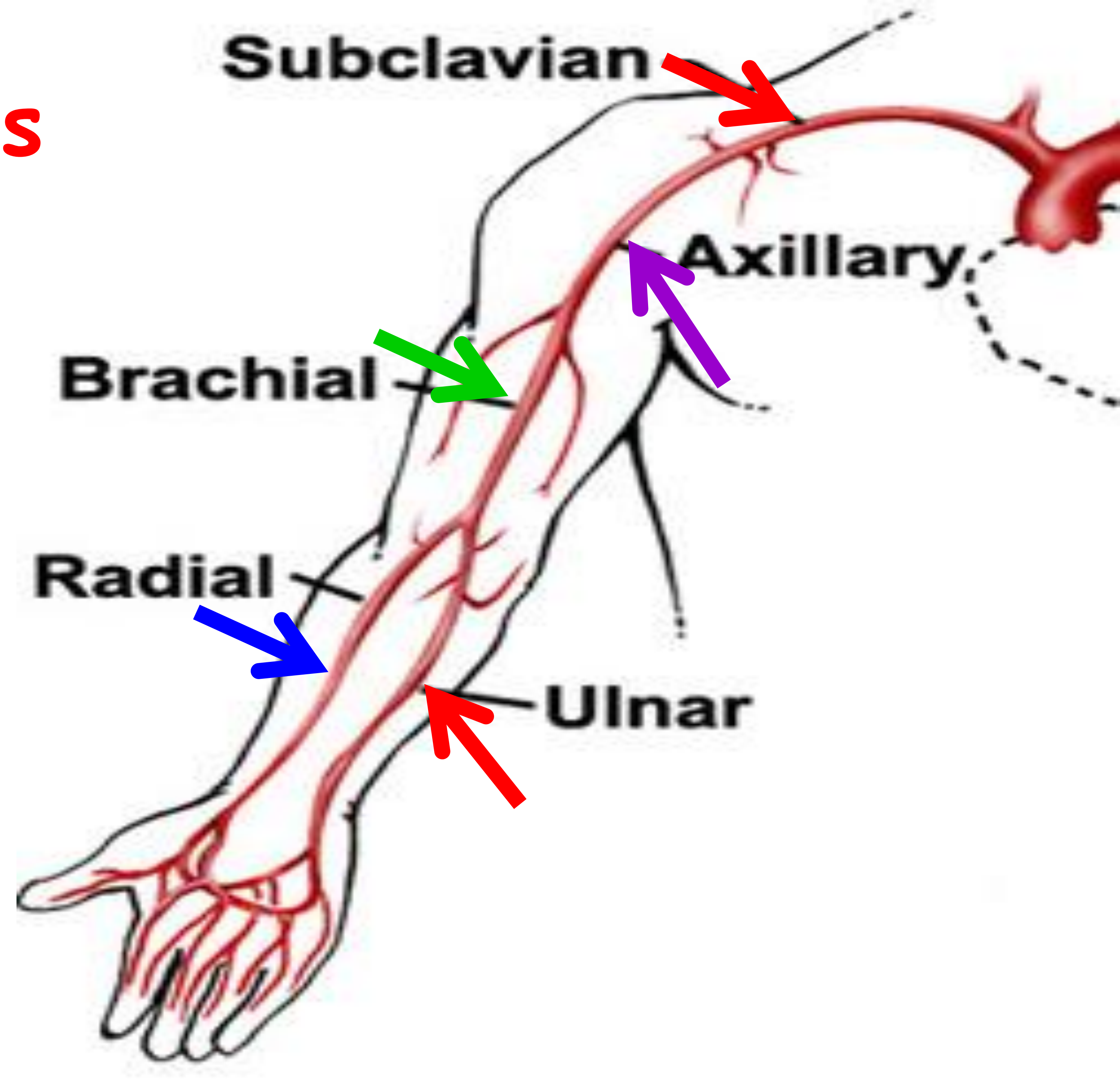
wide, tortuous , lined by
Phagocytes, **present in liver, spleen**
Bone marrow slow down blood and allow exchange

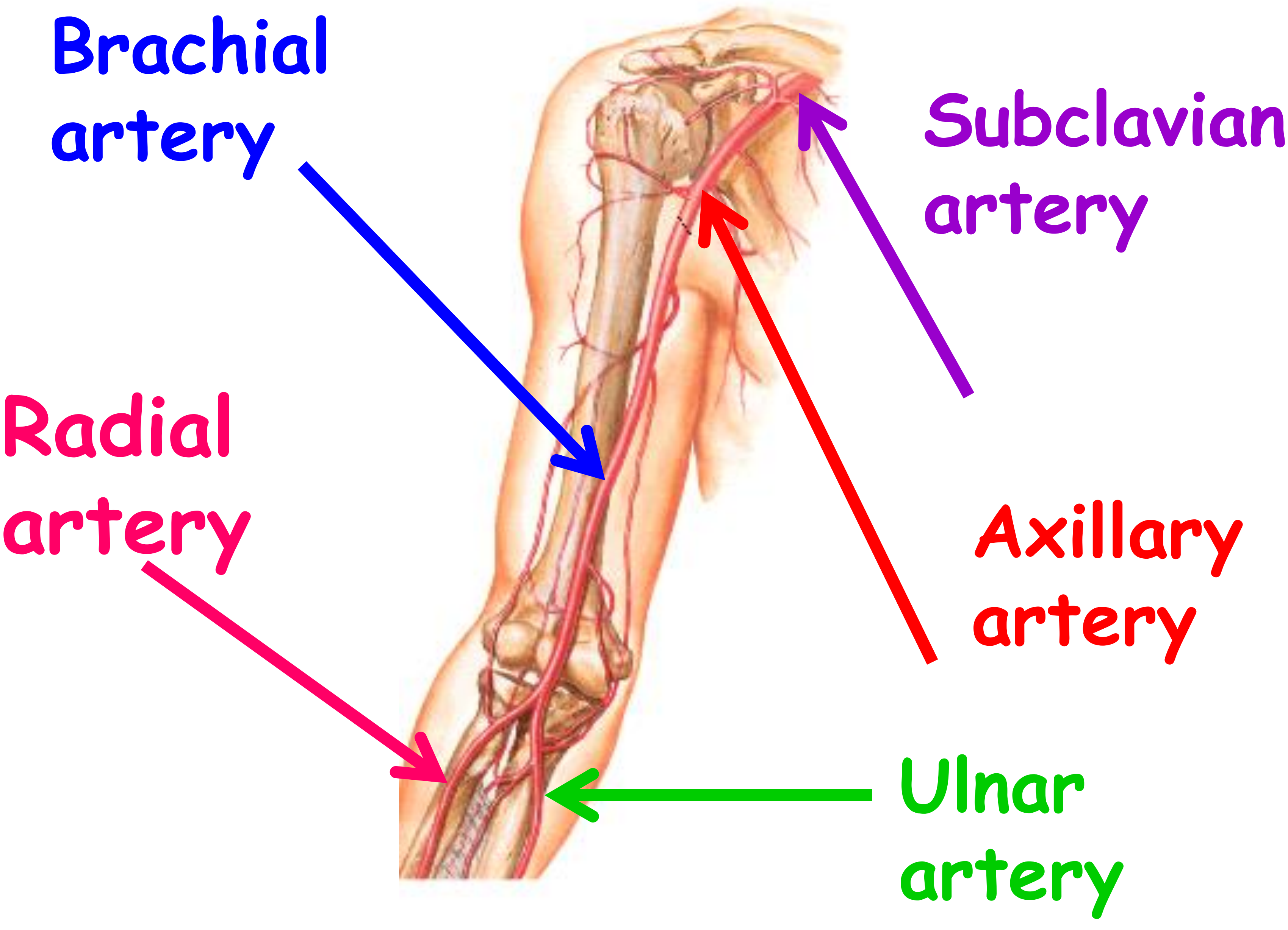
Arteries Of upper limb

Arteries of the upper limb



Arteries of upper limb




Brachial artery
Subclavian artery
Radial artery
Axillary artery
Ulnar artery

VEINS Of upper limb

Veins of the upper limb

A]Superficial veins: *5*

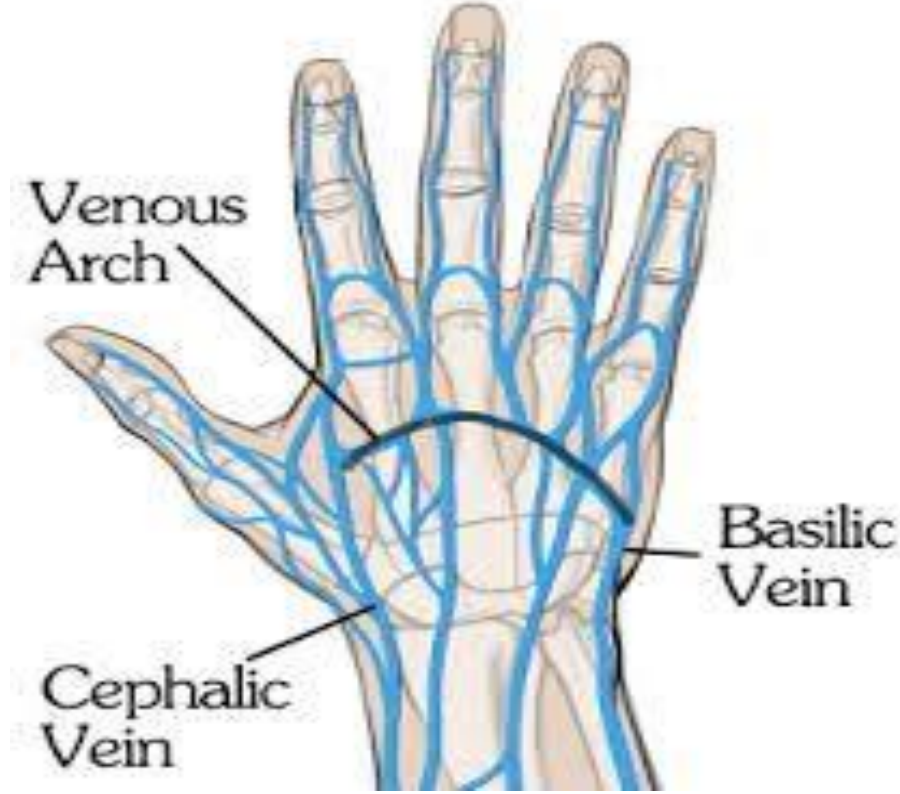
1.Dorsal venous arch.

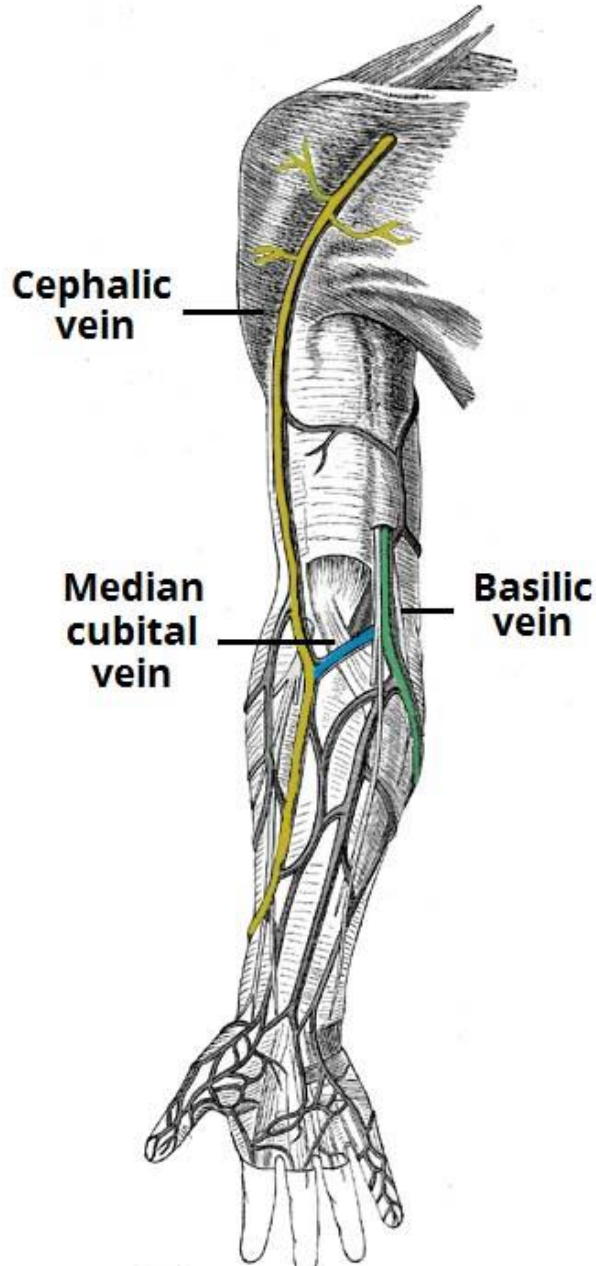
2.Cephalic vein.

3.Basilic vein.

4.Median cubital vein.

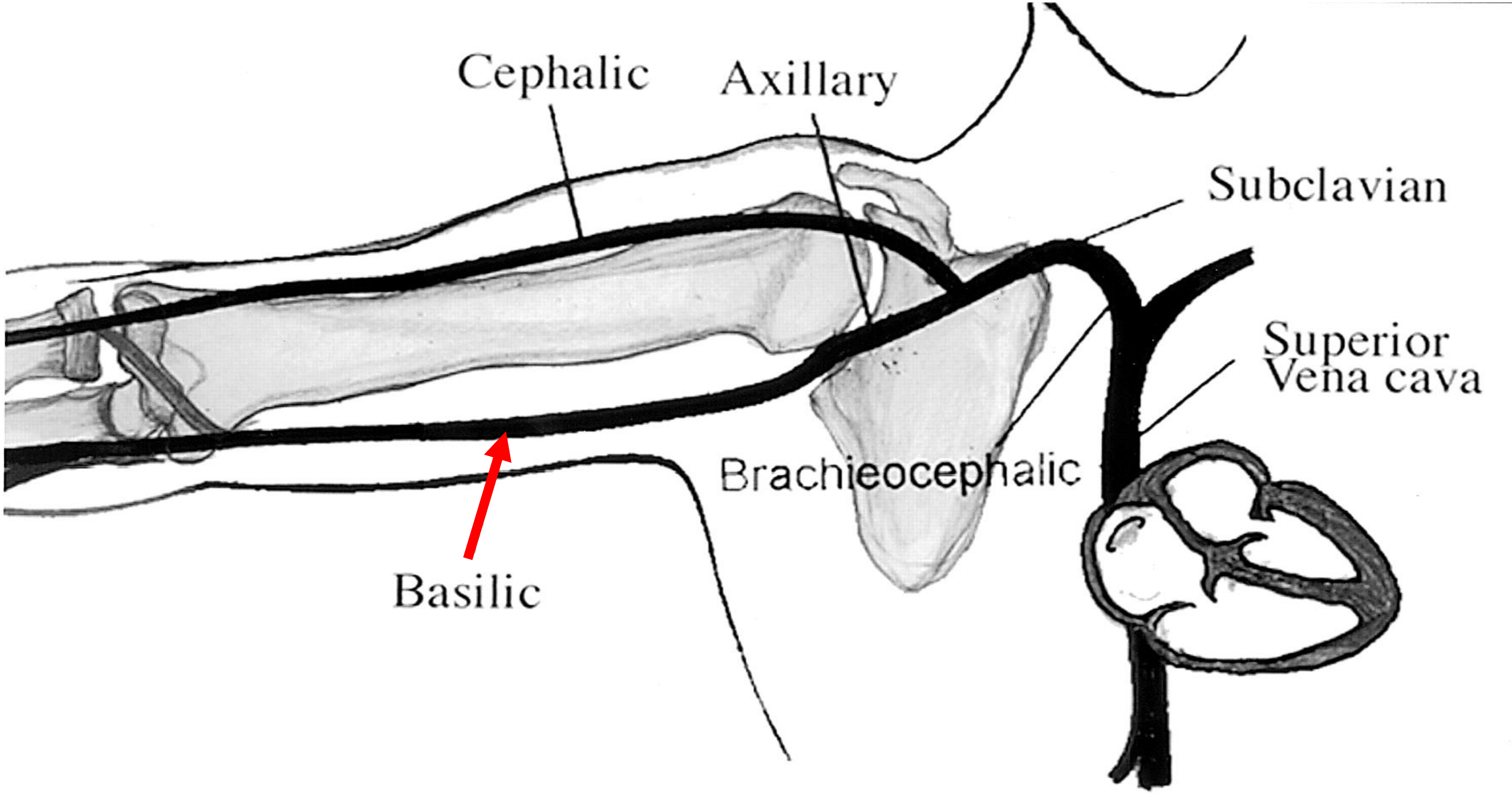
5.Median vein of forearm.





Cephalic vein
Median cubital vein
Basilic vein

Cephalic & basilic veins



Basilic vein (superficial) continues as axillary vein (deep)

Lymphatic System

- Part of circulatory system
- Carries excess fluid that **was not absorbed** by VENOUS SYSTEM
- It consists of

1. **lymphoid tissue**
2. **lymphatic vessels**

Functions of lymph

1. Absorption of excess intestinal fluid
2. Reservoir of fatty elements
3. Antibody production
4. Production of lymphocytes
5. Filtration of bacteria

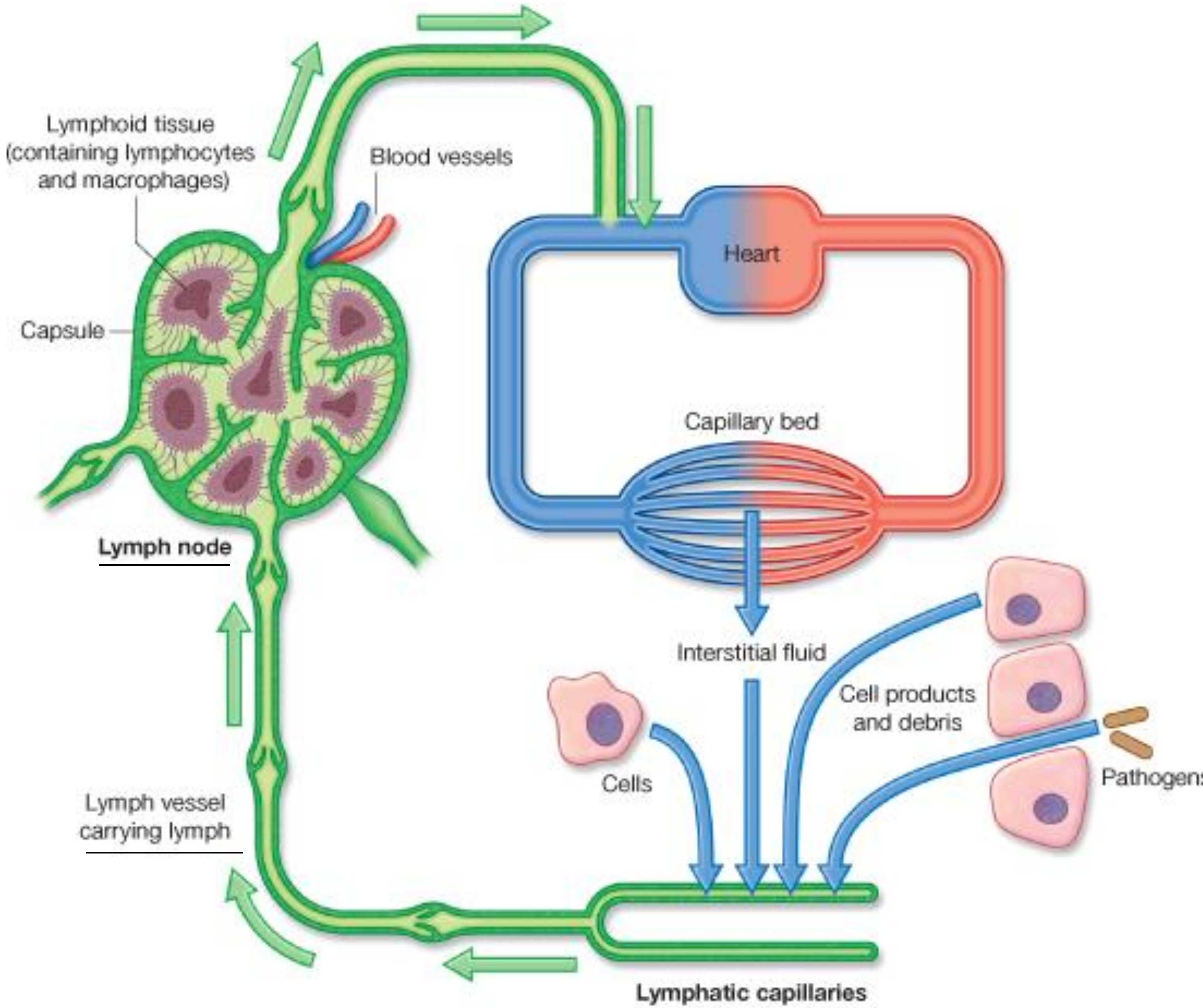
1-Lymphoid Tissue

1. lymph nodes
2. thymus
3. appendix
4. tonsils
5. spleen

2-Lymphatic vessles

- Starts at inter cellular spaces
- Larger& more permeable
- Absorb large molecules and macro organisms
- **Absent in**
- **epidermis, hair nails , cornea**

, brain ,spinal cord , spleen ,bone marrow& internal ear



Lymphoid tissue
(containing lymphocytes
and macrophages)
Blood vessels
Heart
Capsule
Capillary bed
Lymph node
Interstitial fluid
Cell products
and debris
Cells
Pathogens
Lymph vessel
carrying lymph
Lymphatic capillaries

2-Lymph Vessels

- Contain valves & capable of repair
- Either superficial or deep to deep fascia
- **Lymph nodes** run along their course
- **Lymph vessels** carrying lymph to nodes are called **afferents& leaving nodes are called efferents**
- If blocked it causes **edema**
- Inflammation is called **lymphangitis**

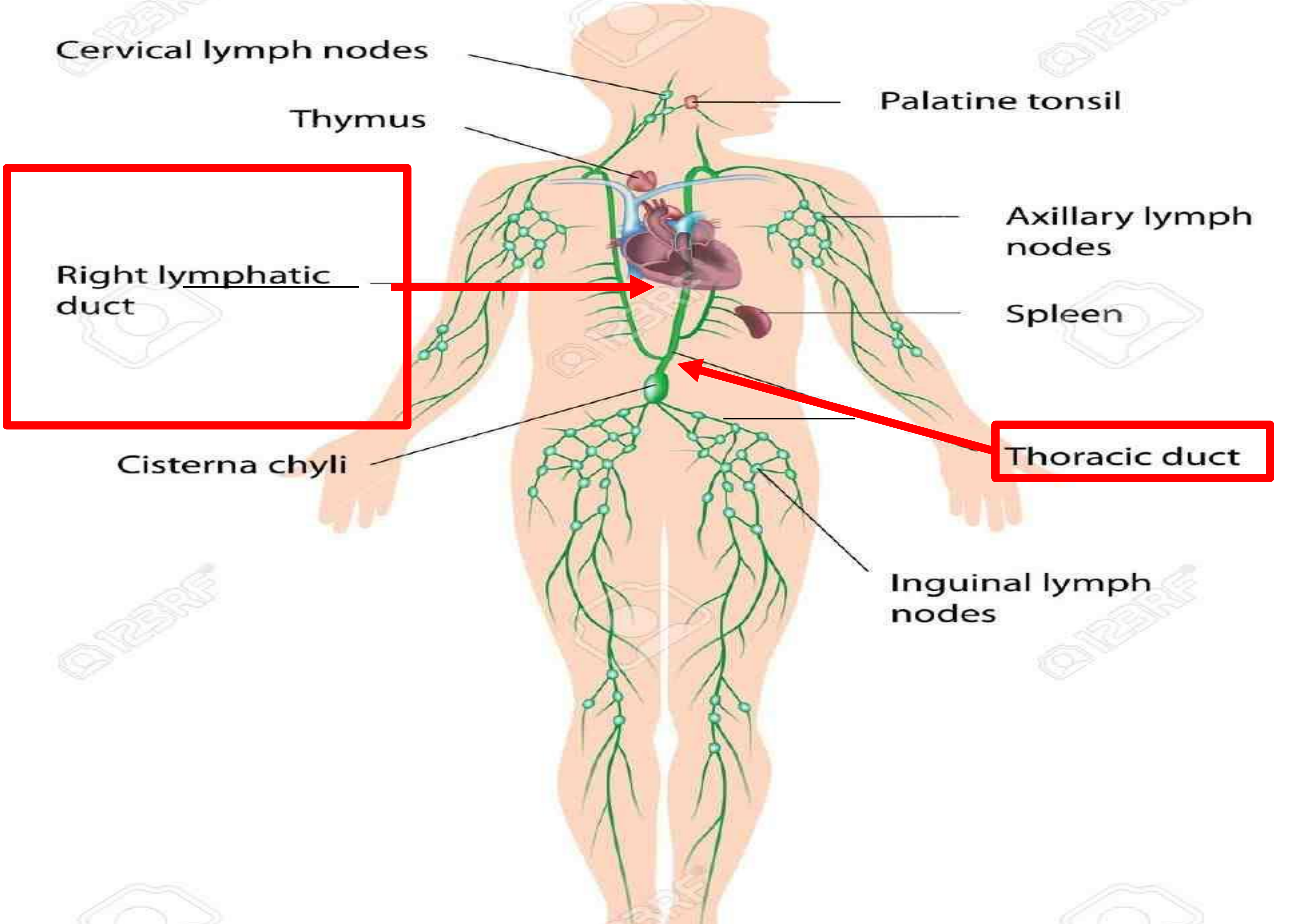
Treatment of Breast Cancer-Related Lymphedema Using

Lymphatic Filariasis –

Ends in Lymphatic Ducts then to veins

- **Thoracic duct**: on left
- **Right lymphatic duct**: on the right

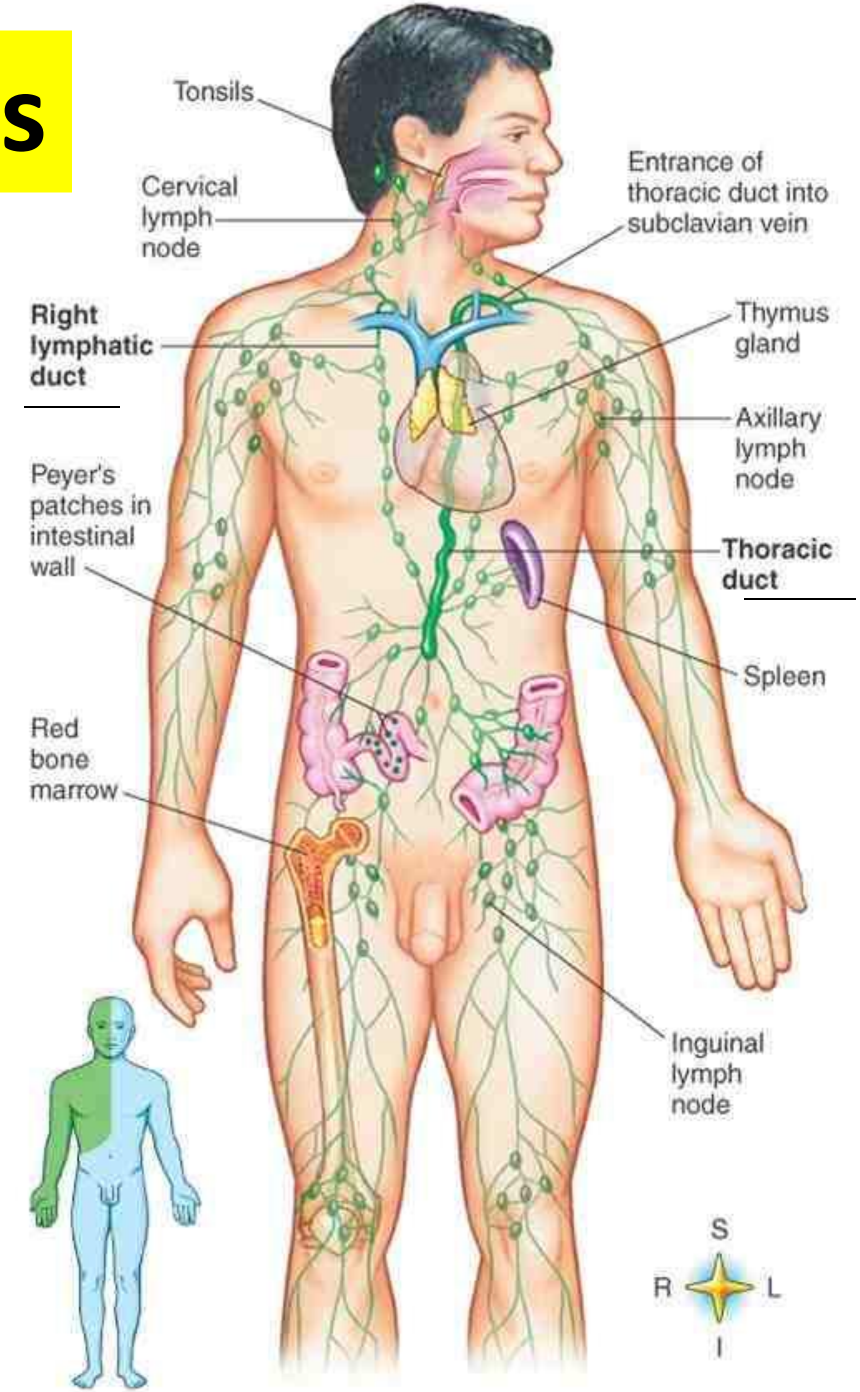
The Lymphatic System



Lymphoid Organs

- **Tonsils**
- **Thymus**
- **Appendix**
- **Spleen**
- **Lymph nodes**

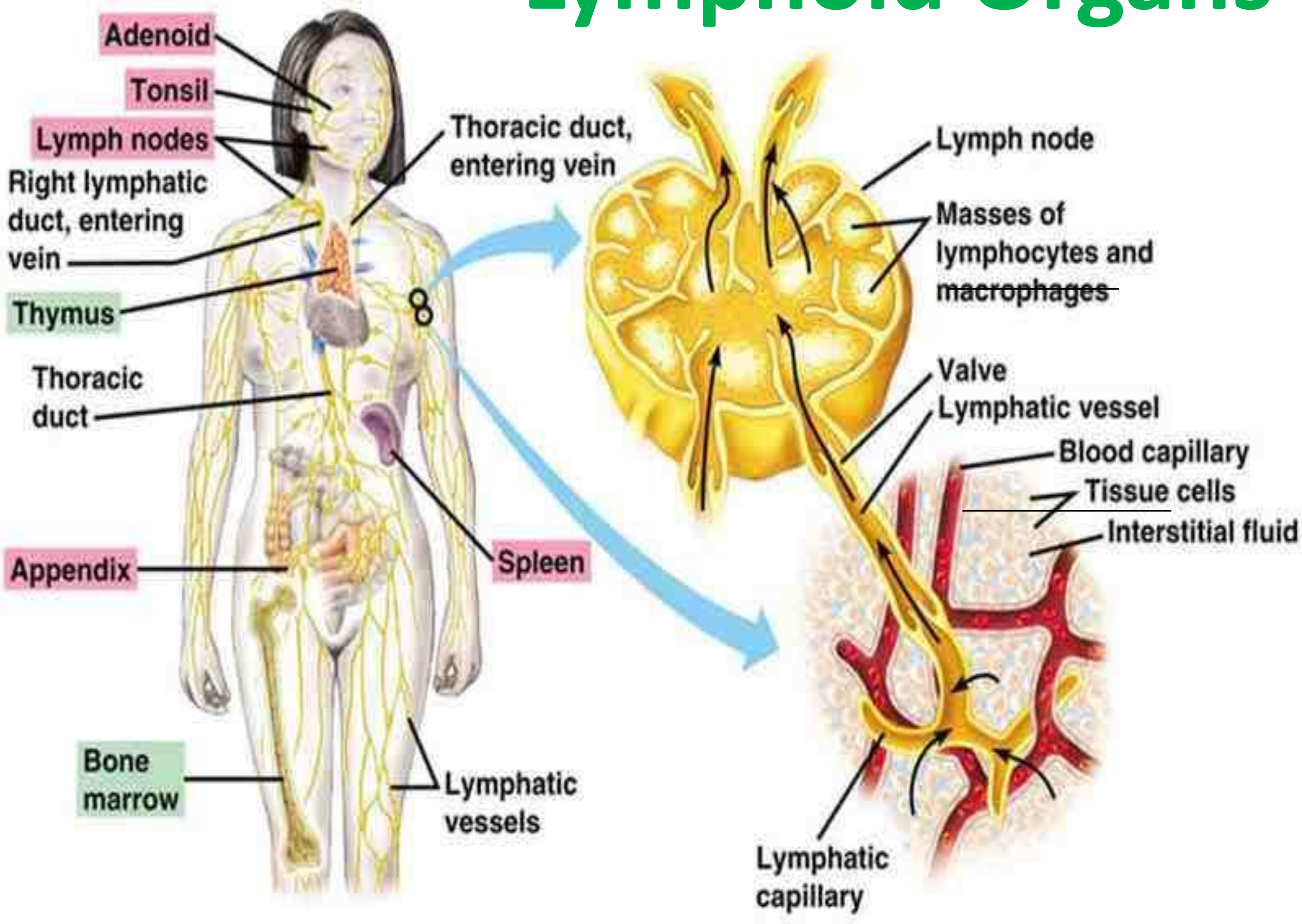
Lymph nodes

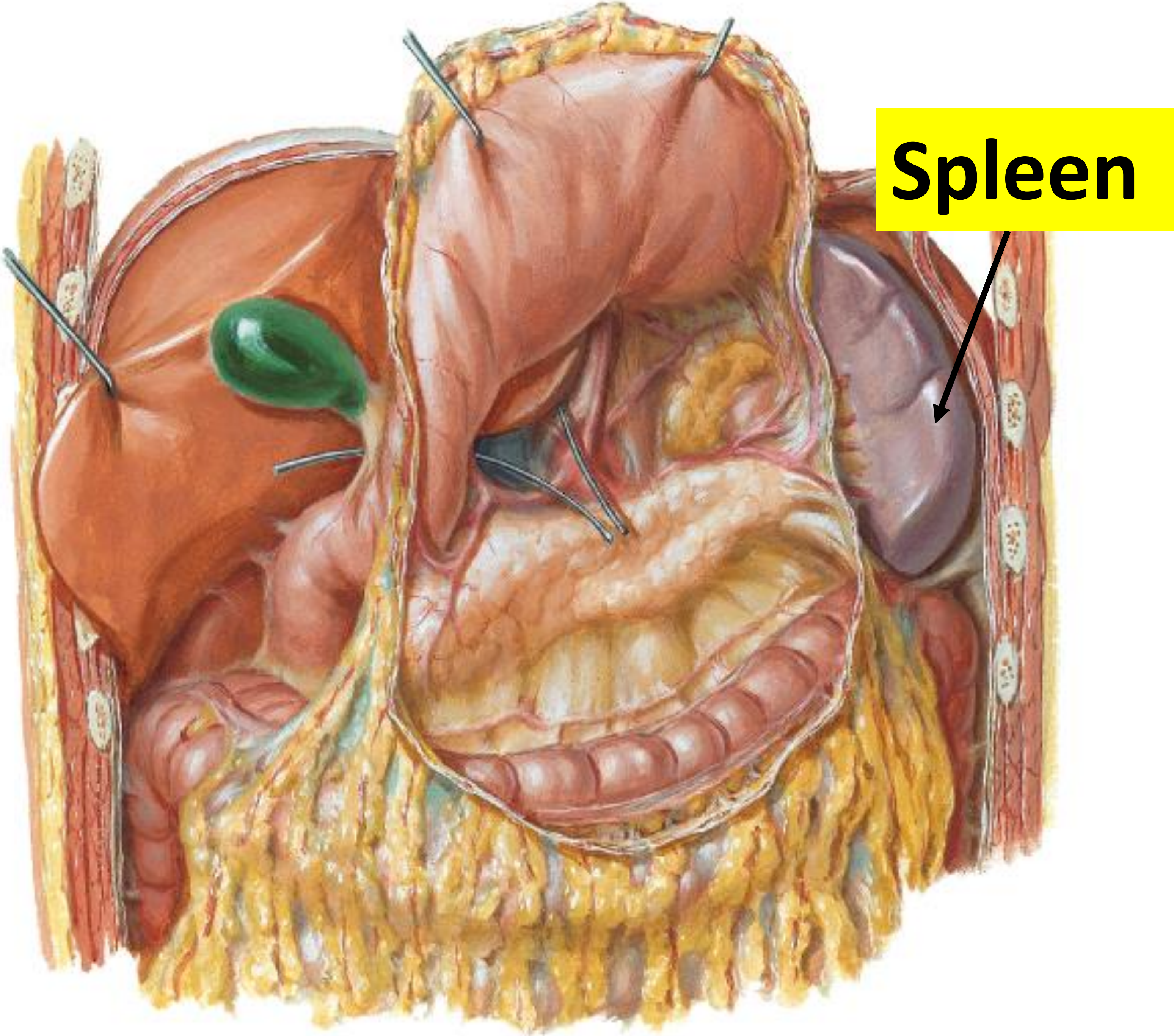


Tonsils
Cervical lymph node
Entrance of thoracic duct into subclavian vein
Right lymphatic duct
Thymus gland
Axillary lymph node
Peyer's patches in intestinal wall
Thoracic duct
Spleen
Red bone marrow
Inguinal lymph node
S
R
L
I

Lymph nodes

Lymphoid Organs




Spleen

Anatomy of Respiratory System

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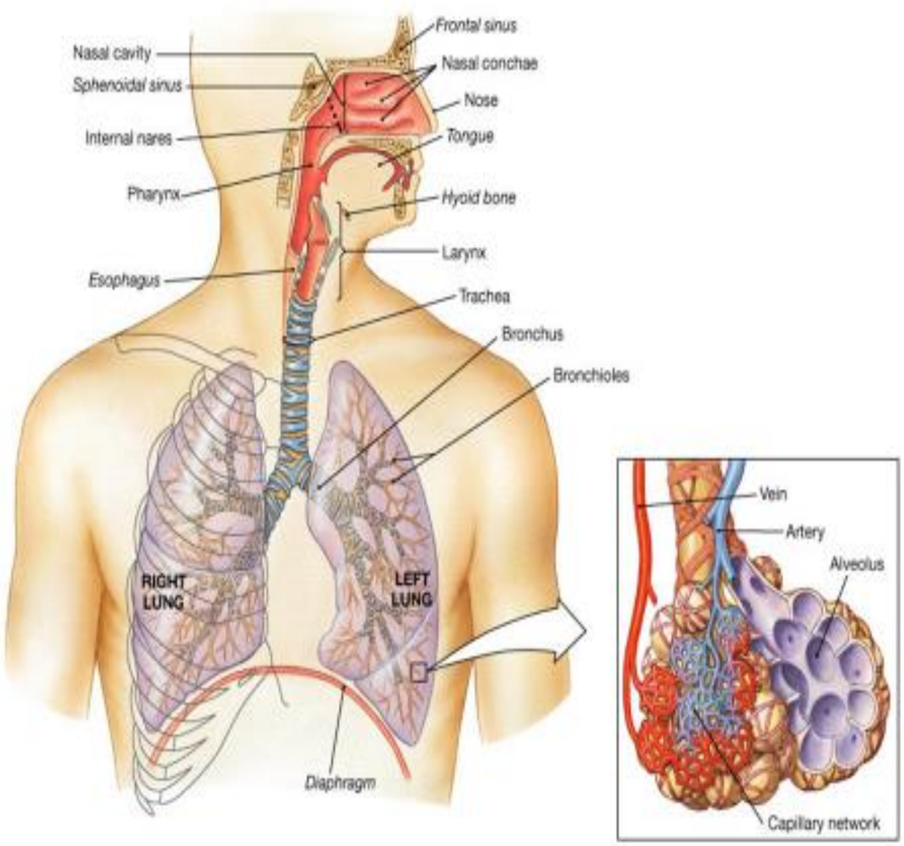
Anatomy of Respiratory System

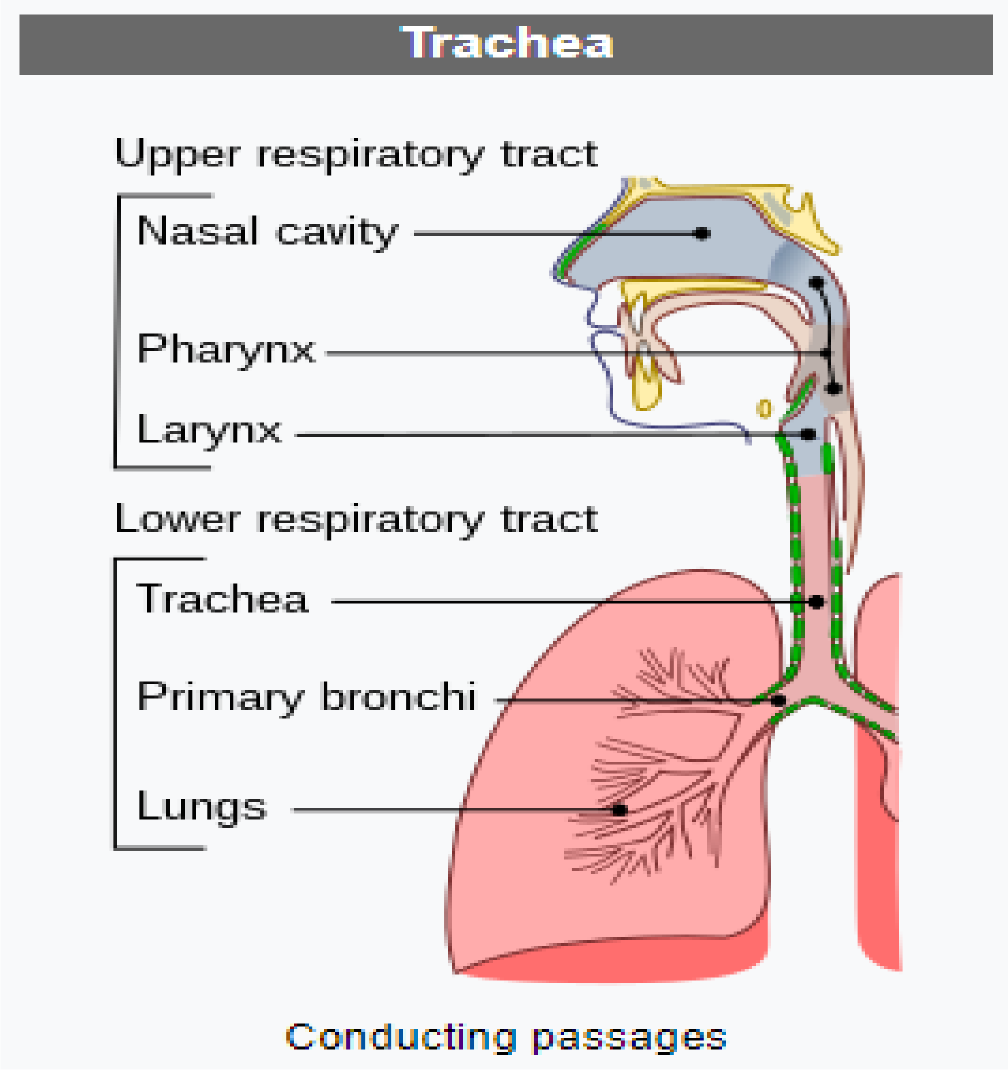
- <u>**The respiratory system formed of**</u>
- **Nose**
- **nasal cavity**
- **pharynx**
- **larynx**
- **trachea**
- **Bronchi**
- **lungs**



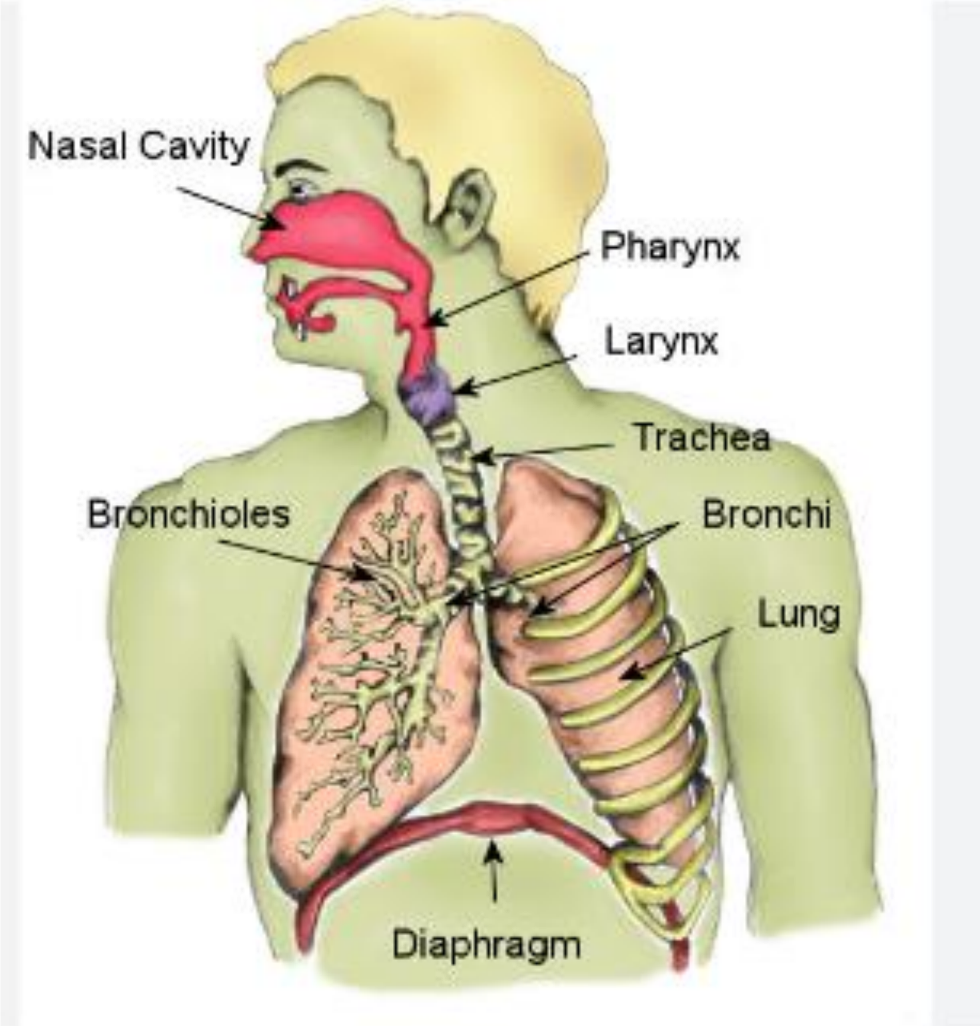
It is Consists of :

1. upper respiratory tract (nose to larynx)

1. lower respiratory tract (trachea downwards).



- <u>1-Conducting portion transports air.</u>
- includes the nose, nasal cavity, pharynx, larynx, trachea, and progressively smaller airways, from the primary bronchi to the terminal bronchioles



- <u>2-Respiratory portion carries out gas exchange.</u>
- - composed of small airways called called alveoli

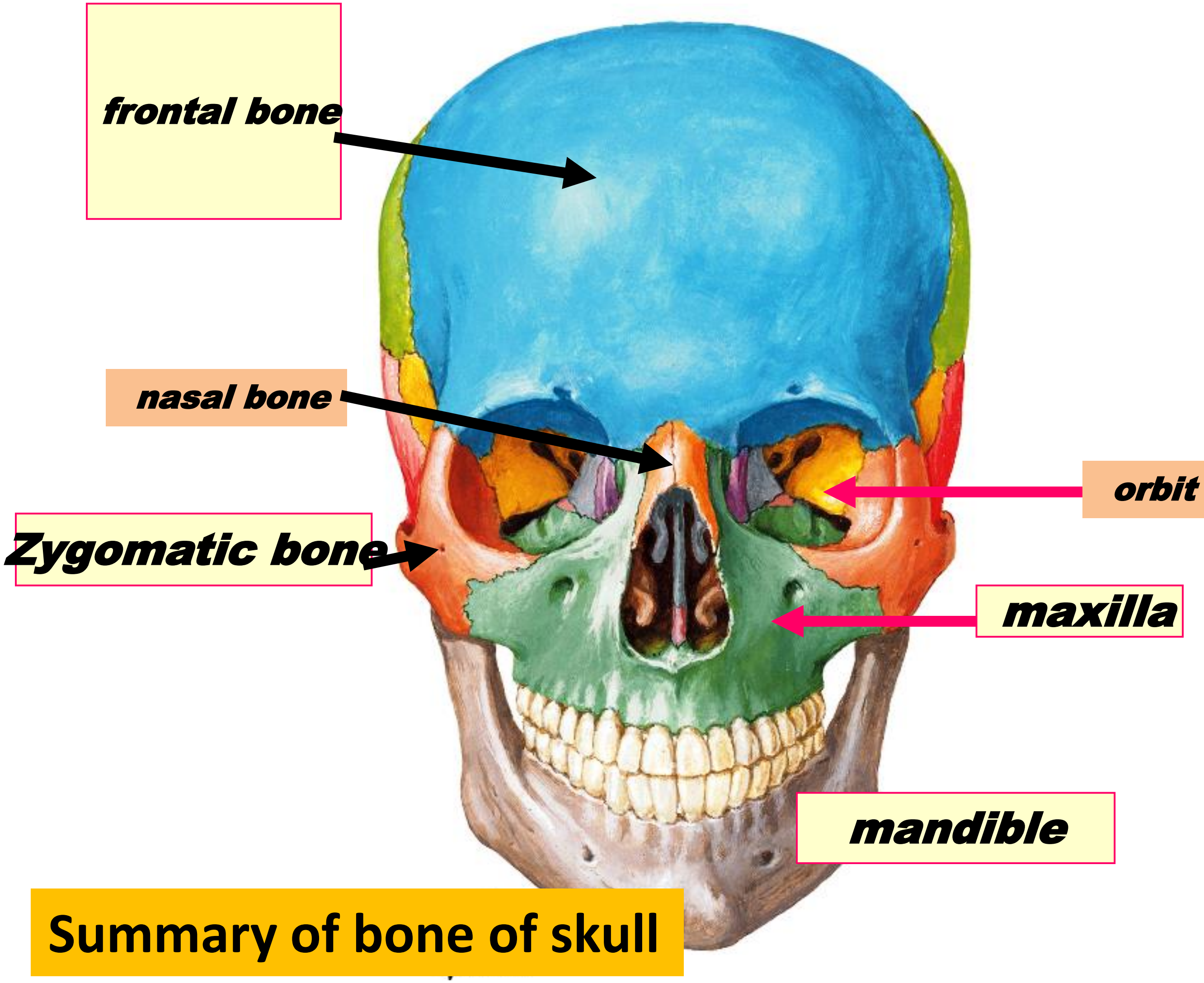
1-NOSE:

- it is formed **of 2 nasal cavities** separate by a **nasal septum**
- It is important to
  1. **moistens and warms enterin air**
  2. **filters and cleans inspired air**
  3. **resonating chamber for speech**
  4. **detects odors in the air stream**
  5. **provides and airway for respiration**

rhinoplasty

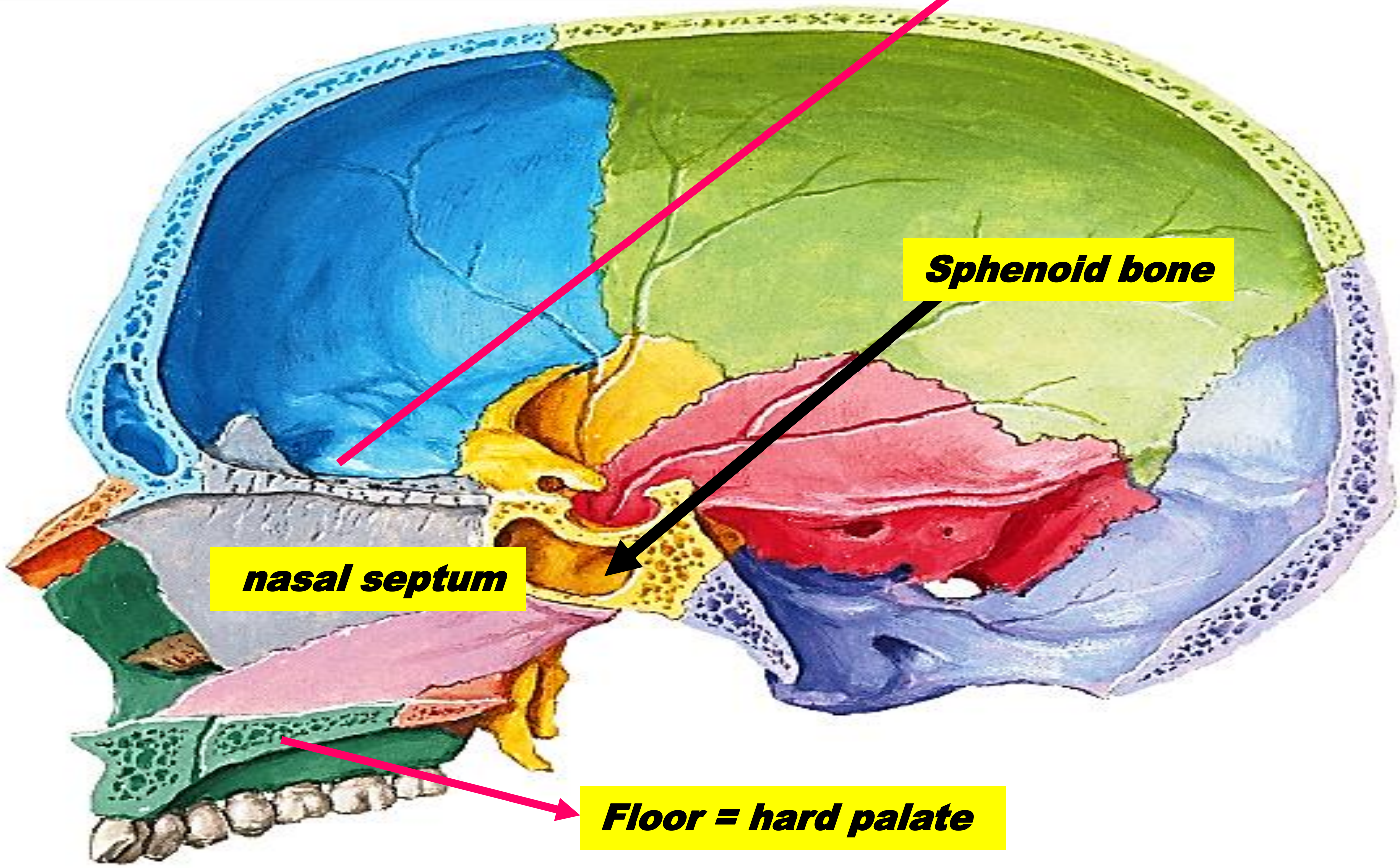
surgery to change shape of external nose

Para nasal sinuses:



Summary of bone of skull

The Nasal Cavity



Para nasal sinuses:

- **Definition : are spaces inside skull filled with air**
- **to lighten weight of skull & air conditioning .**

Sinuses are named according to the bones they are located in.

- **they include**
- **1- Frontal air sinus**: in frontal bone
- **2- Sphenoidal air sinus**: in sphenoid bone of skull
- **3- Ethmoidal air sinus:** between orbit & nose
- **4- Maxillary air sinus** inside maxilla

Headache DUE TO sinusitis ??

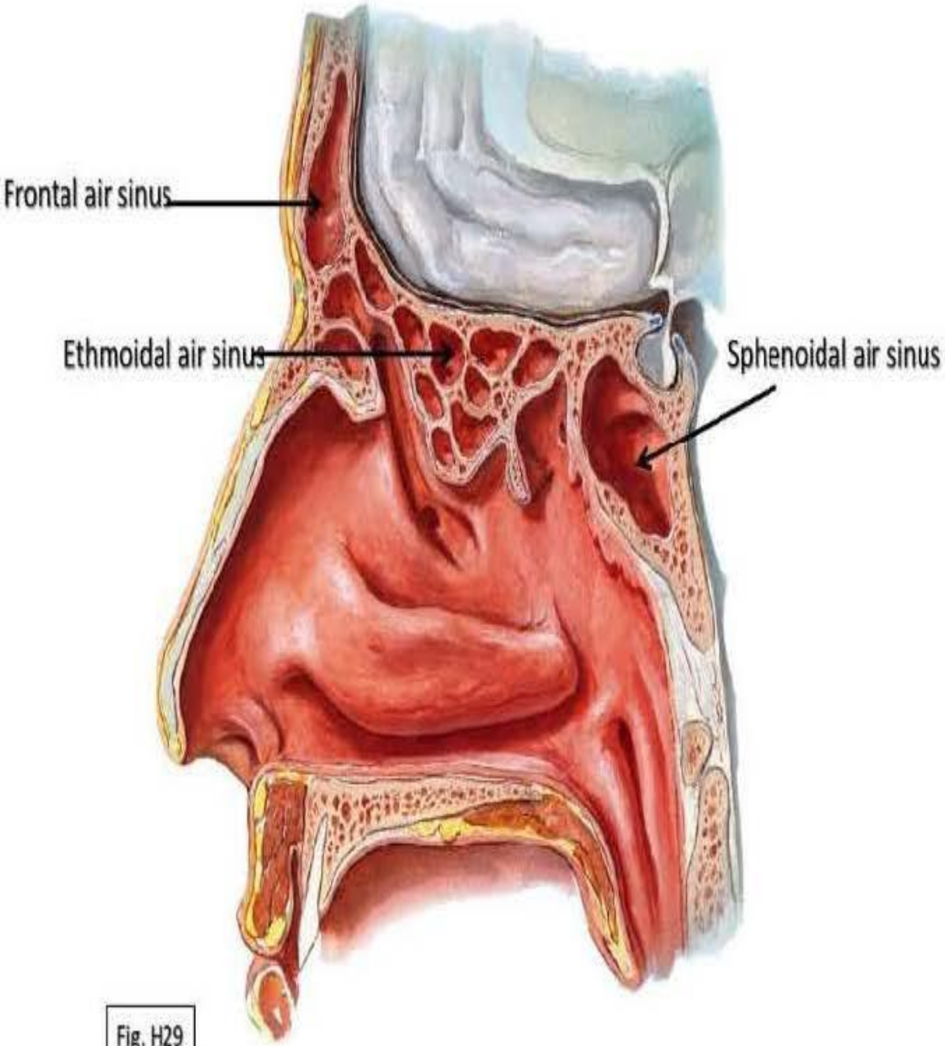


Fig. H29

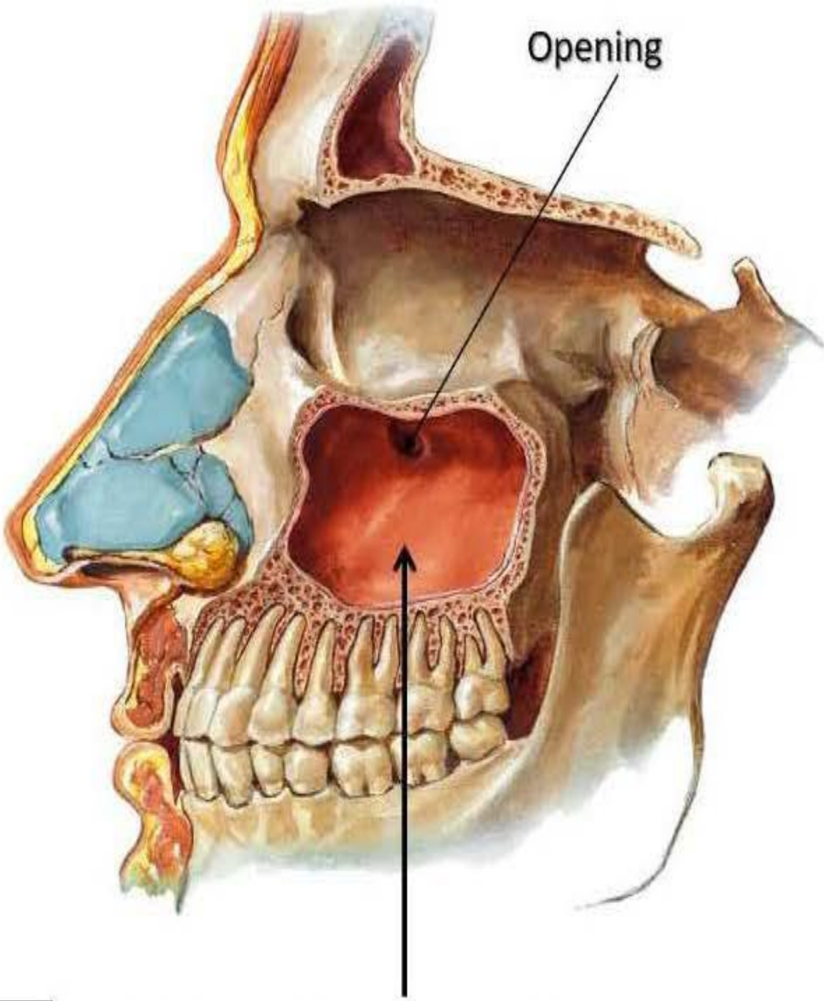
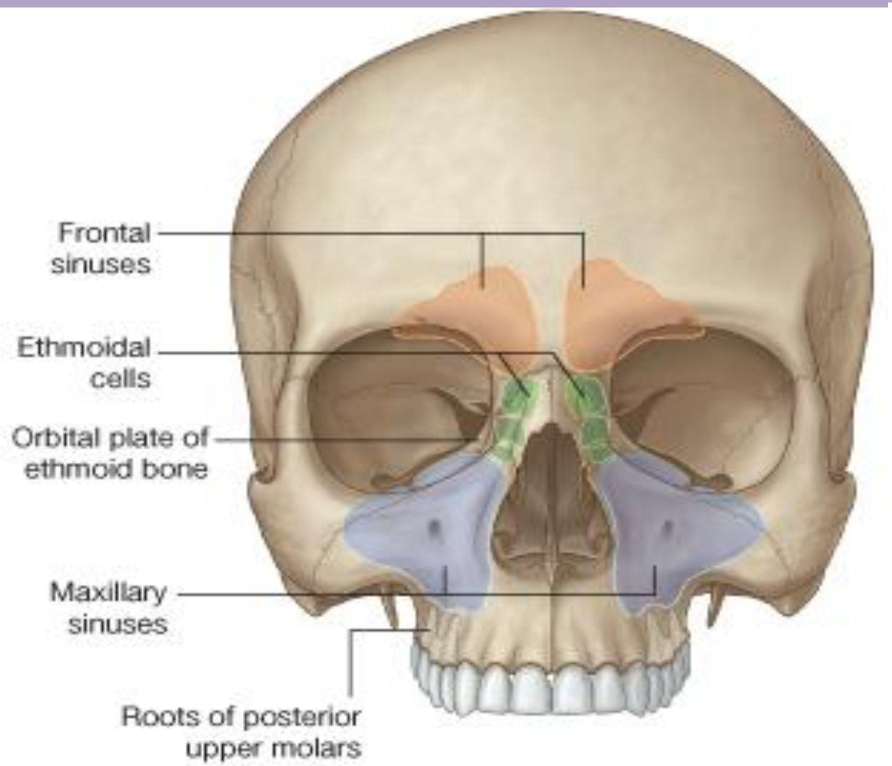
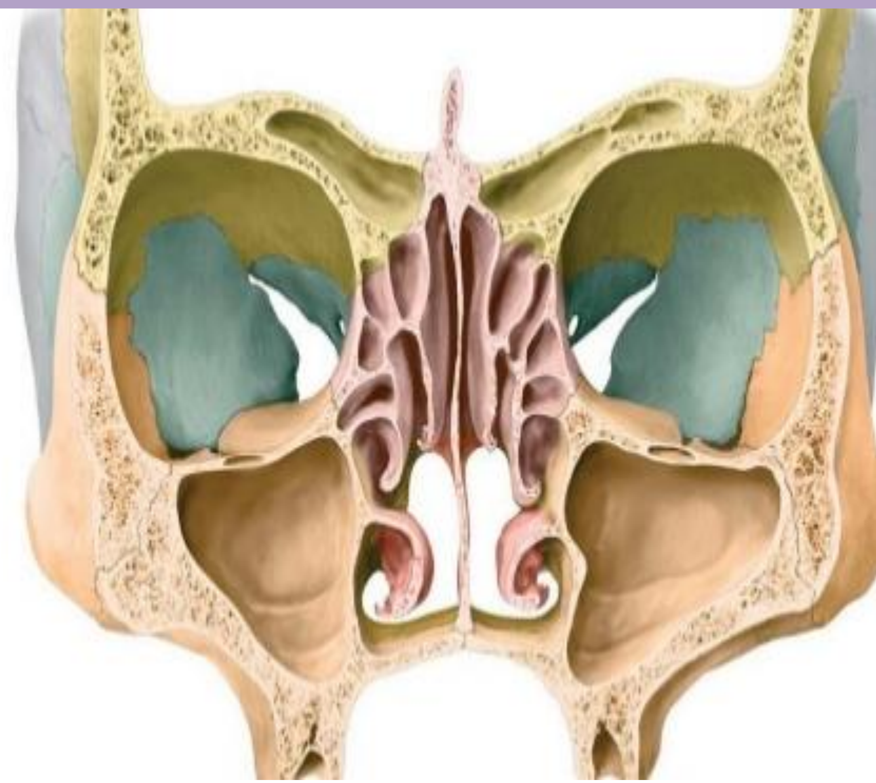


Fig. H30

Paranasal Air Sinuses

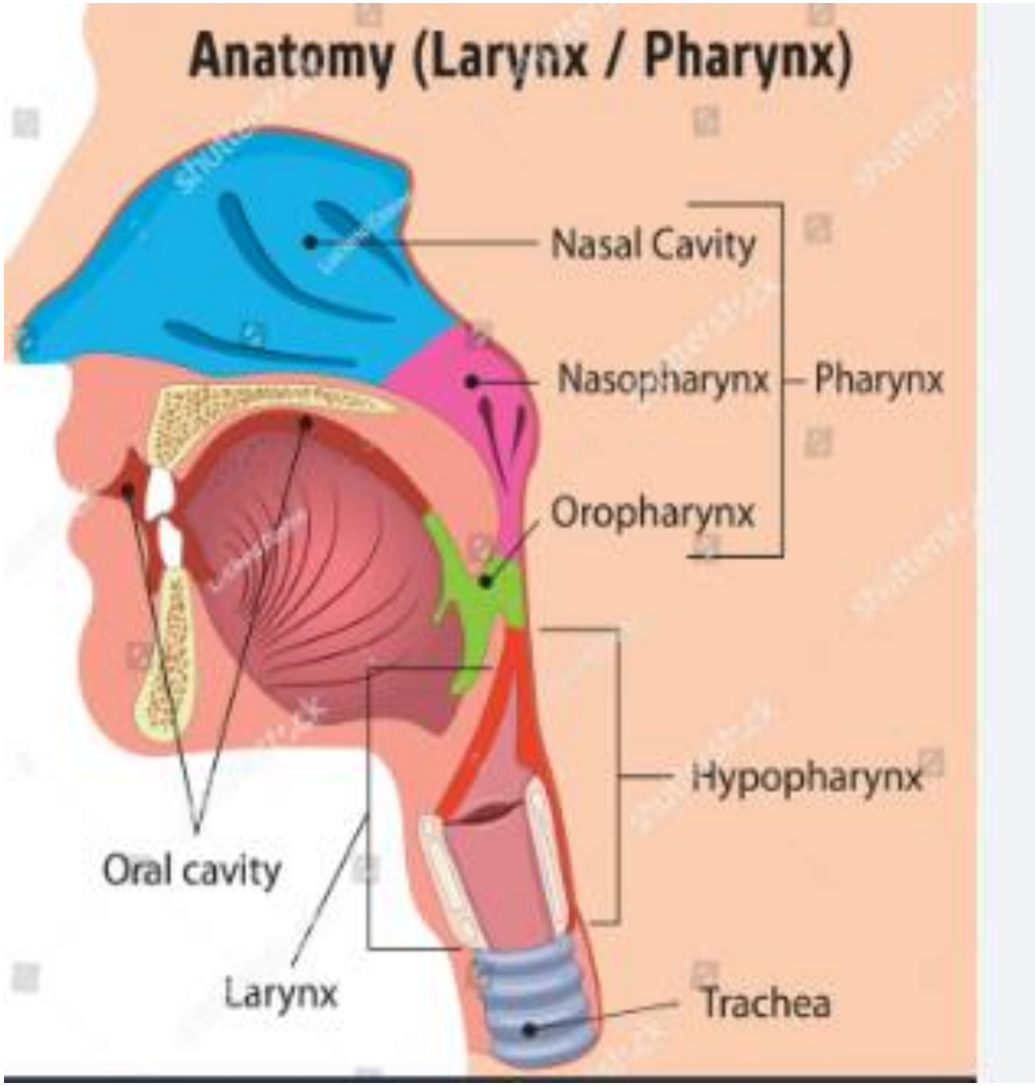
<u>Definition</u>: These are air-filled spaces located inside the skull bones around the nasal cavity.





2-Pharynx

- Common space used by both the respiratory and digestive systems. Common pathway for both air and food
- Commonly called the throat.
- It is a tube lies behind the nose, mouth and larynx

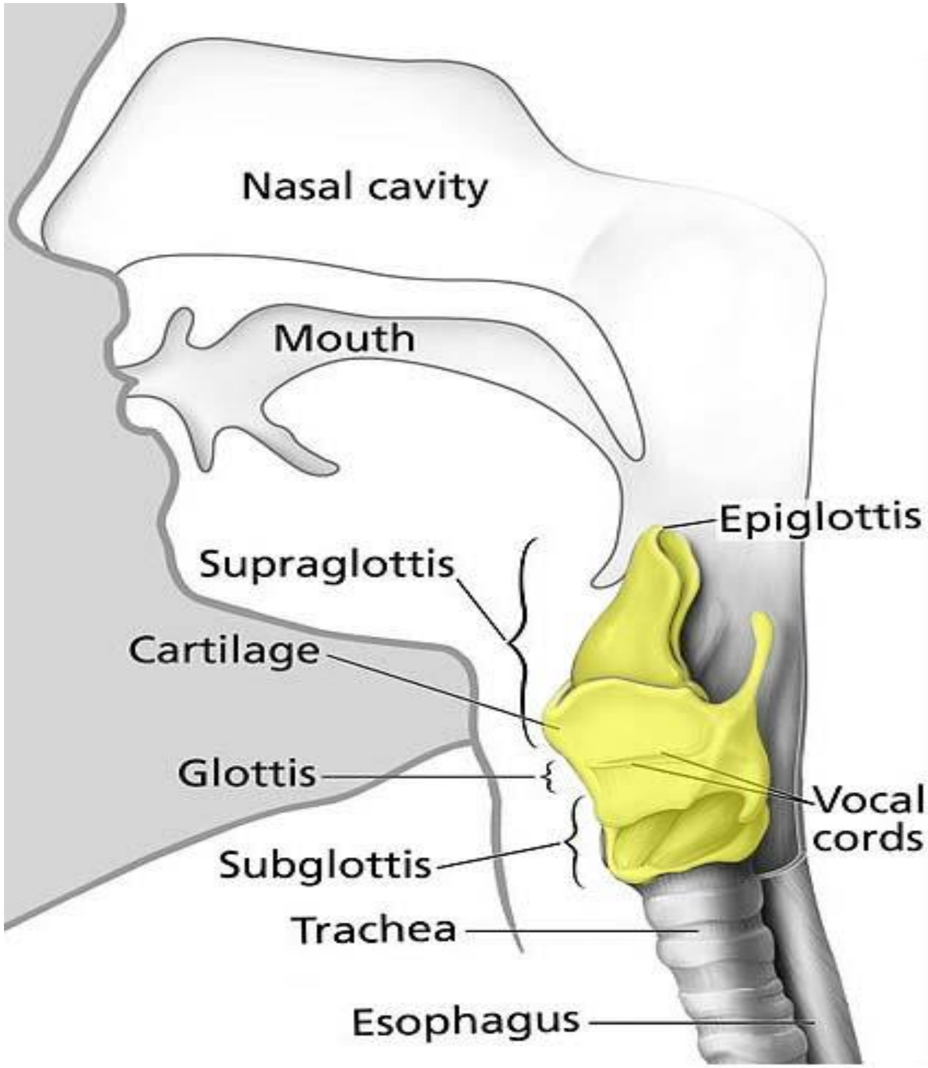


3- Larynx Voice box

1. It is a tube lies in the neck
2. ends in the trachea
3. formed of 9 cartilages.

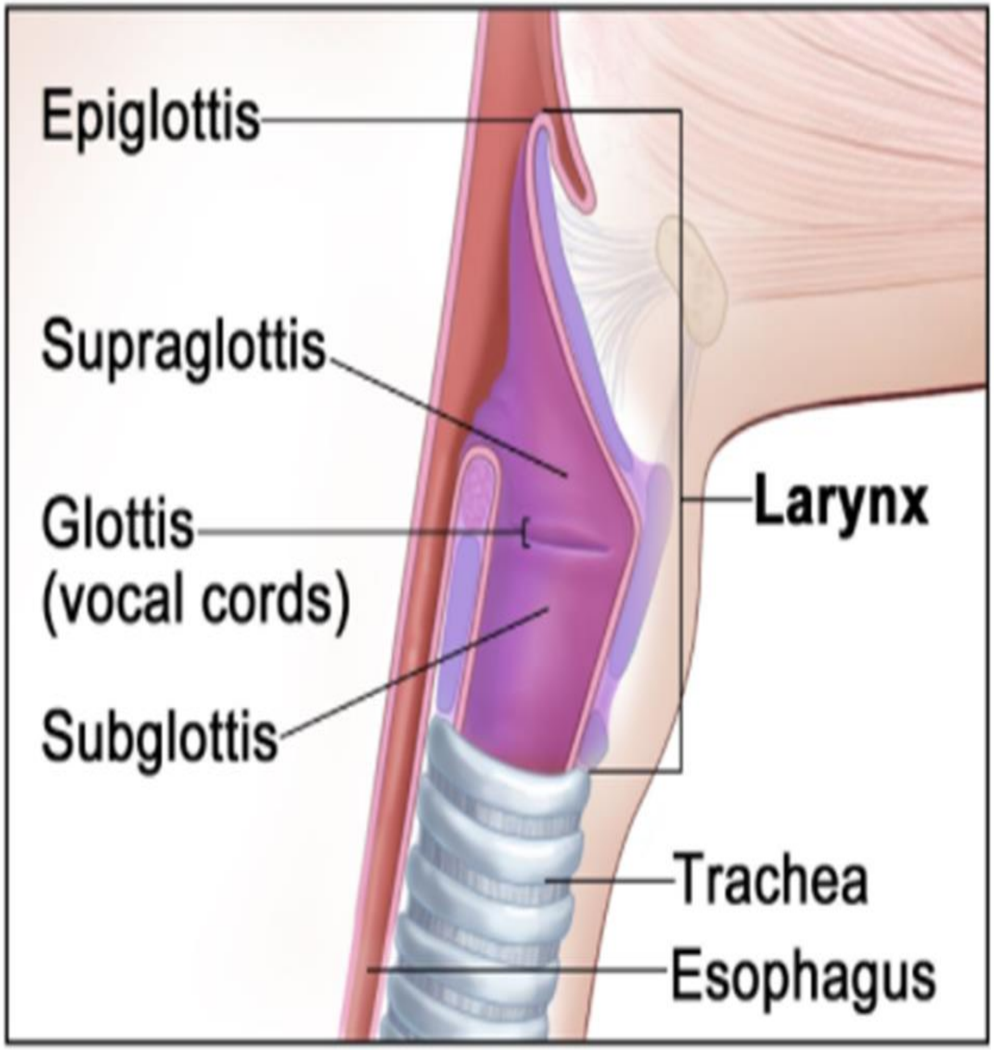
■ It is closed by epiglottis.

■ It Prevents swallowed materials from entering the lower respiratory tract.



3- Larynx Voice box important in

- Conducts air .
- speech



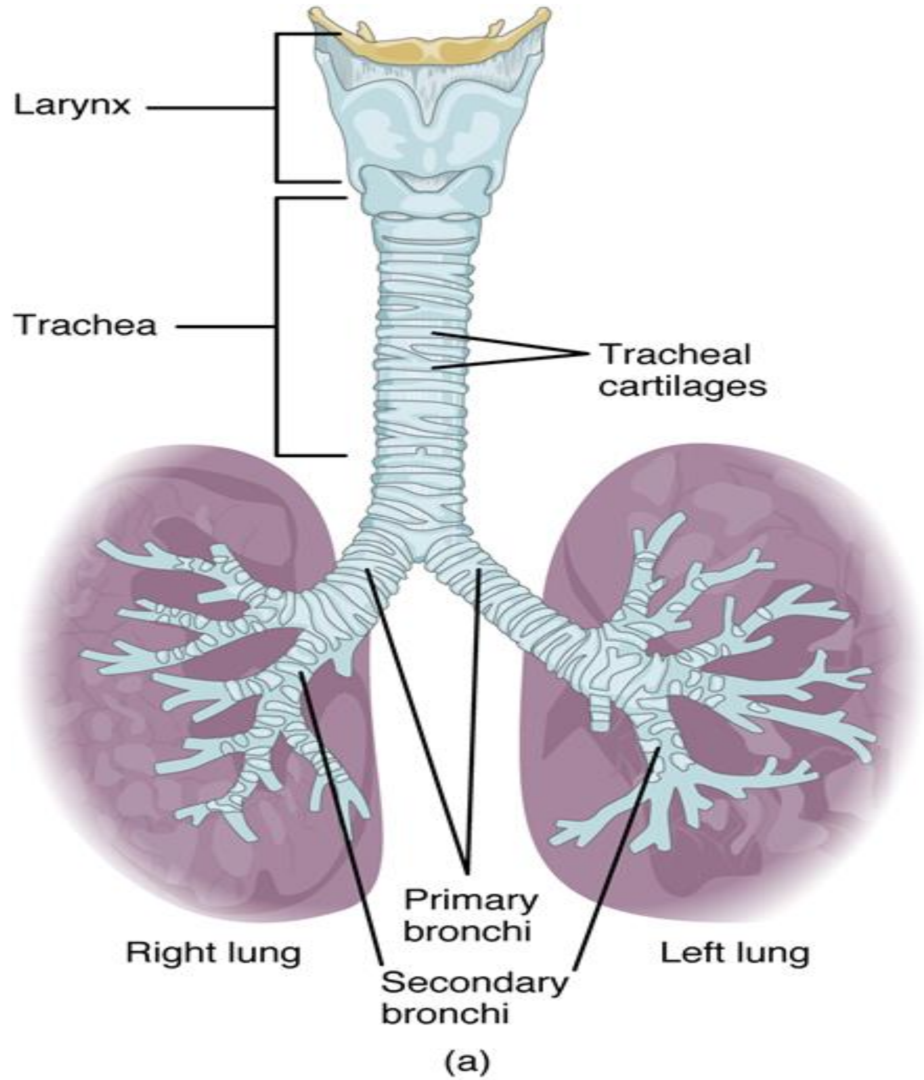
4-Trachea

- It is a tube =20 CM
- formed of C shaped cartilages

- that begins in the neck at larynx
- ends in the thorax.

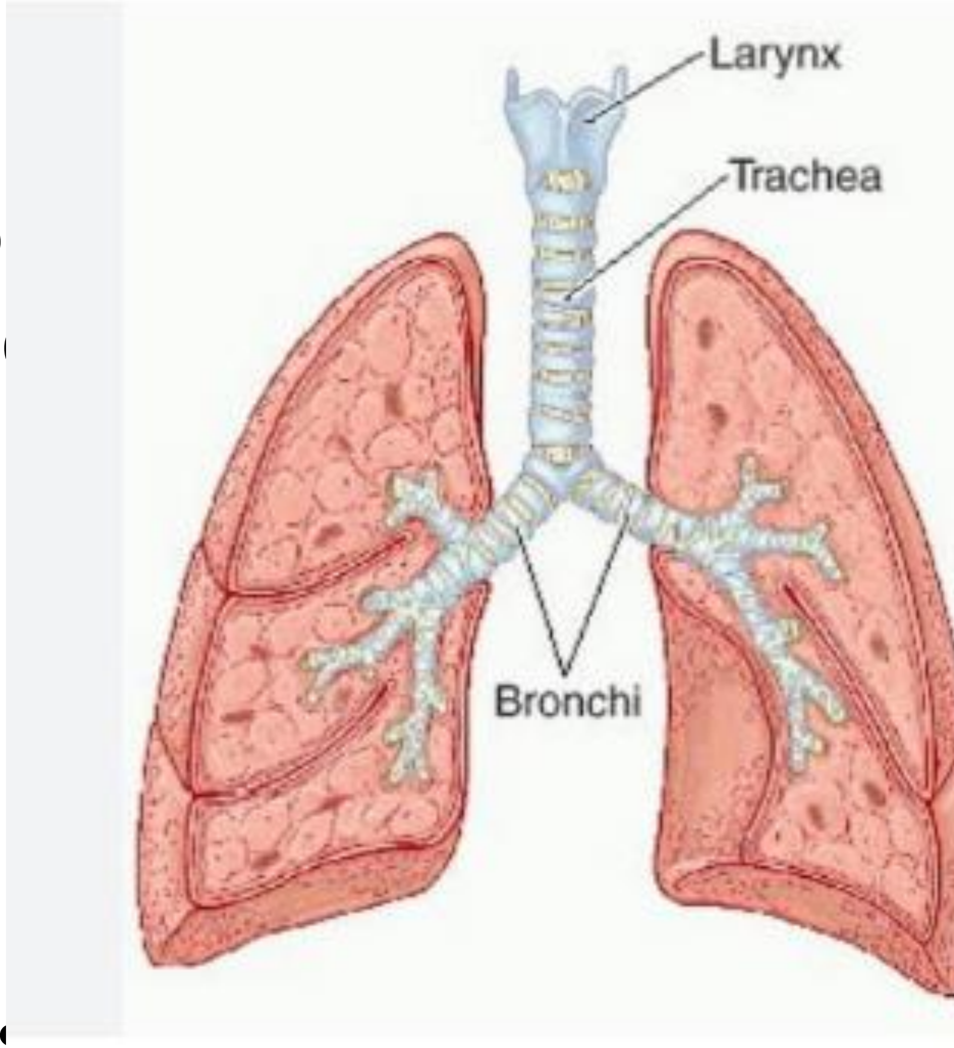
- divides into two tubes, called the right and left bronchi.

- Each bronchus enter the lung.



5-Bronchi

- **the trachea** divides into two smaller tubes, called the **right and left bronchi.**



1. the **right** bronchus enters the **right** lung
2. **the left bronchi** enters the **left** lung.

Right bronchus

1. Right bronchus is <u>shorter, wider, and more vertically in line with the trachea</u>
2. Foreign body are more likely to pass from the trachea to the right bronchus.

6-The lungs

- Shape: a conical. with apex up and base down
- sit : Lungs lie in the thorax on each side of the heart
- Its wide, concave base rests on the diaphragm.
- the apex projects superiorly the clavicle.
- heart
- 2 surfaces (costal and

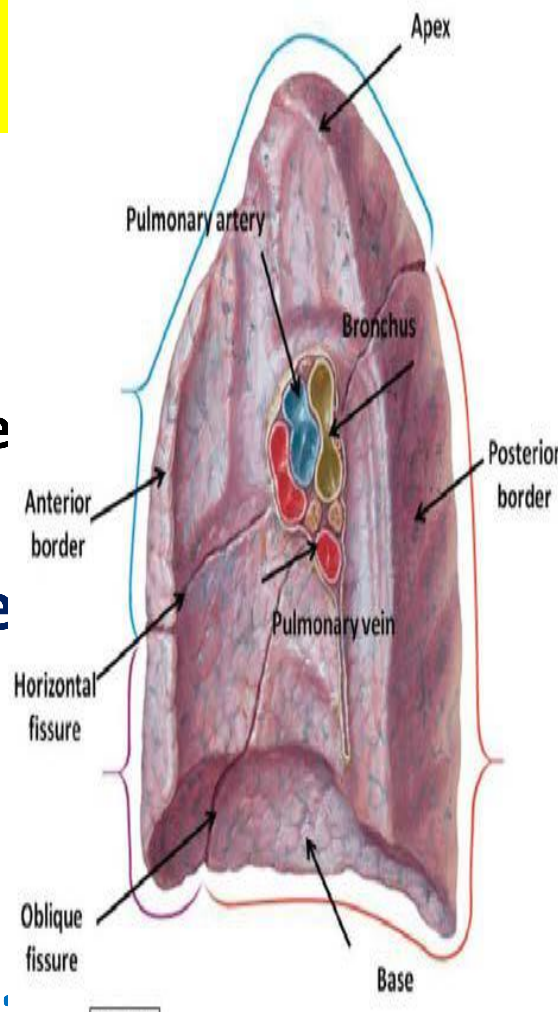


Fig. T15

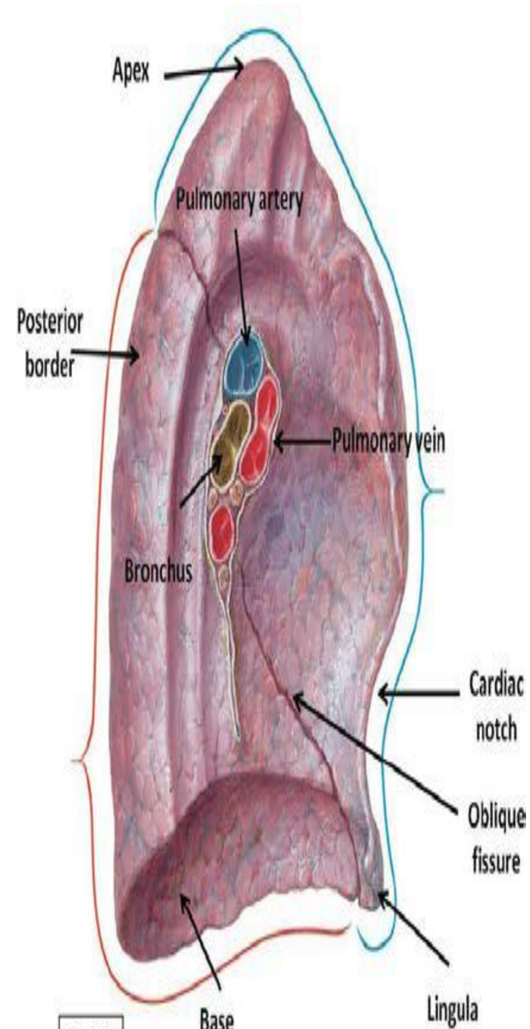
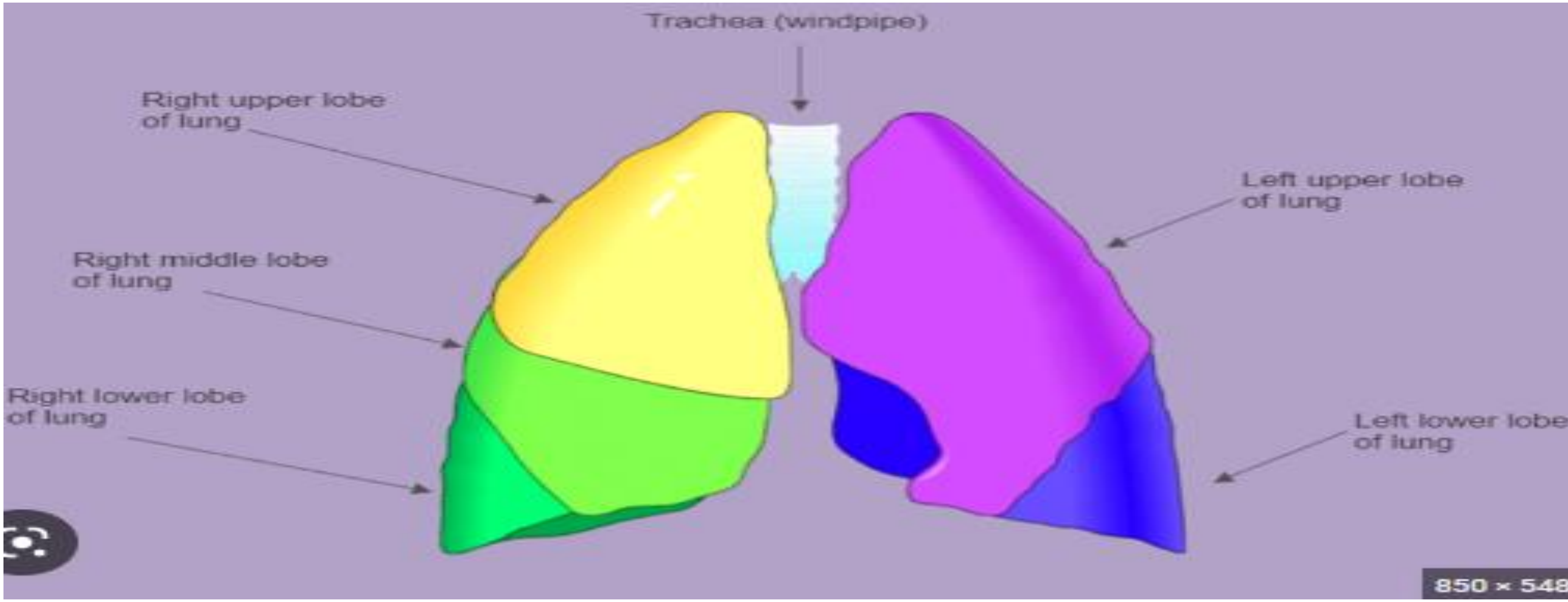


Fig. T16

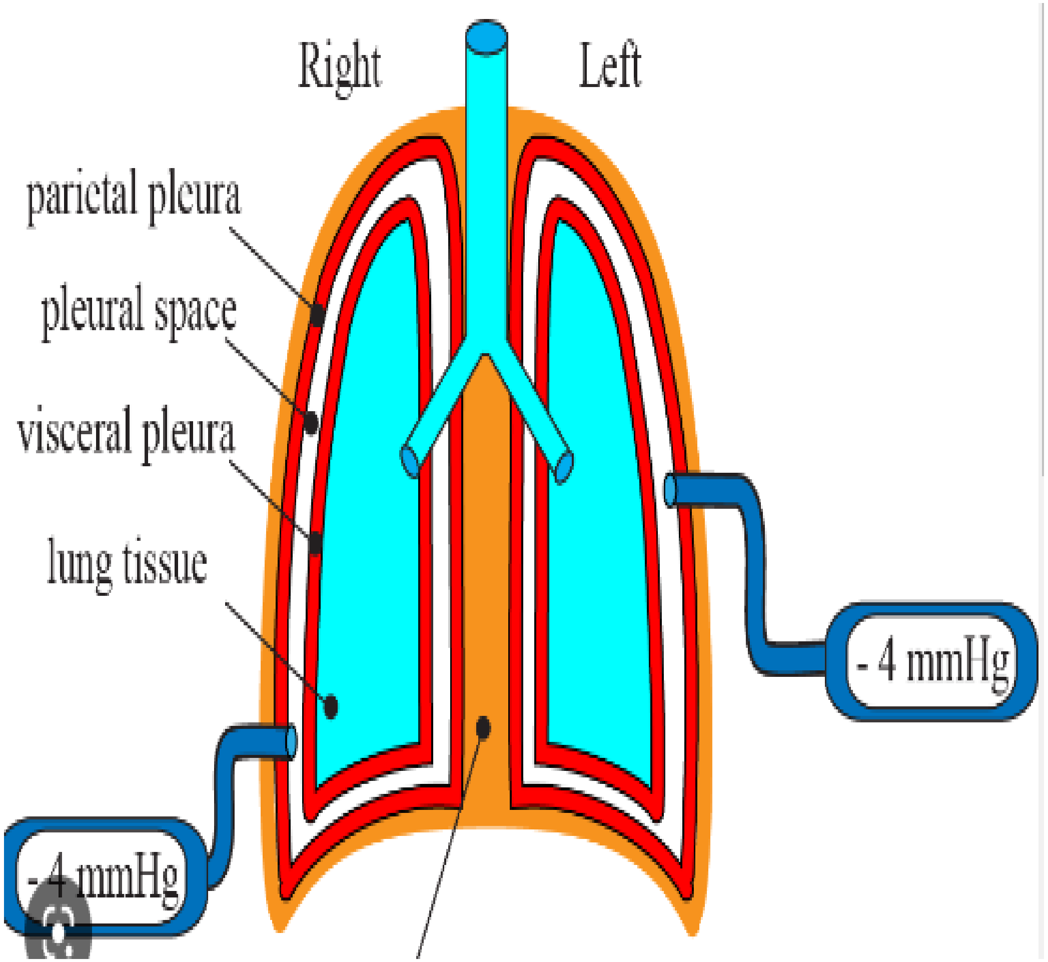


- **Right lung**
- **divided into 3 lobes by oblique and horizontal fissure**
- **more superiorly due to liver**

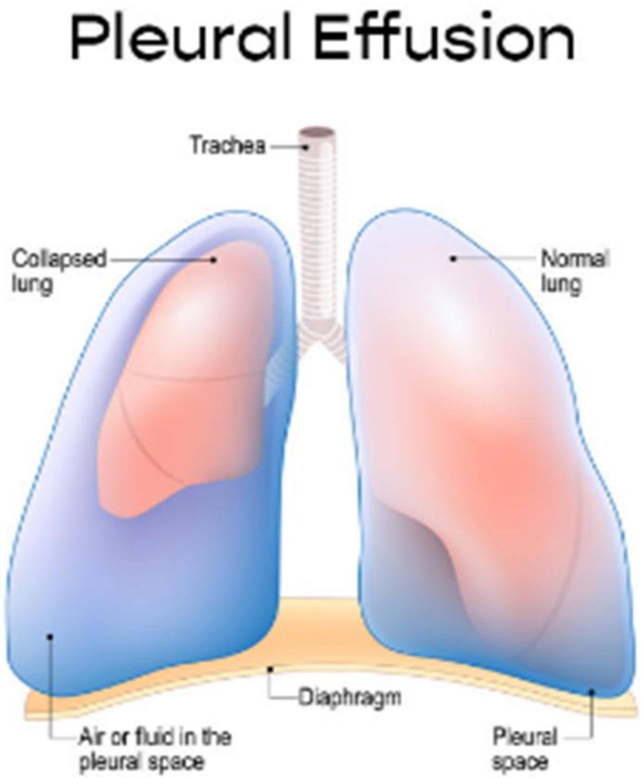
- **Left lung**
- **divided into 2 lobes by oblique fissure**
- **smaller than the right lung**
- **cardiac notch for the heart**

7-Pleura

- The lung surrounded by a serous membrane called **pleura.**
- It is formed of 2 layers (visceral **pleura**& **parietal pleura) and pleural cavity in between.**

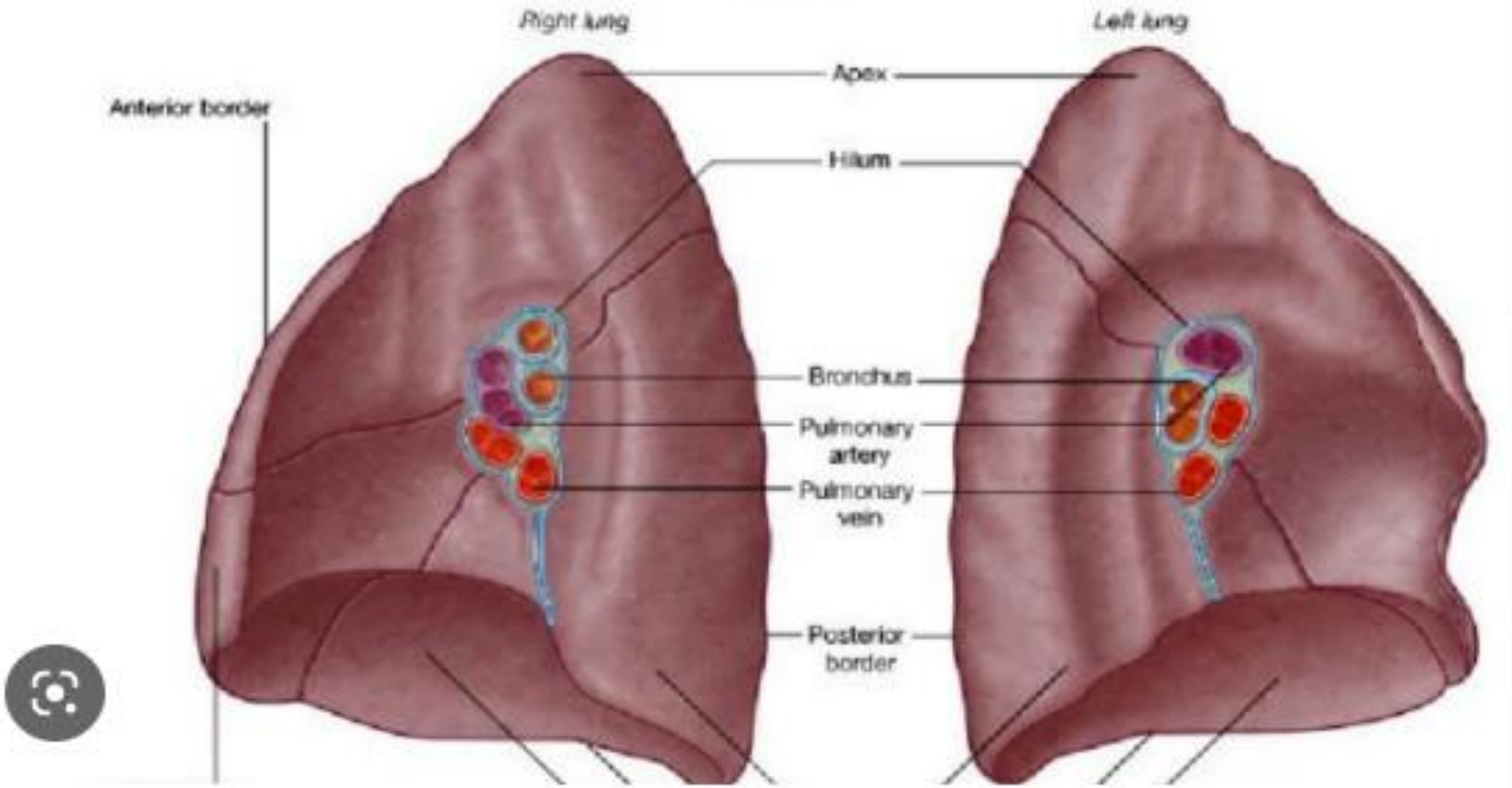


- **pleural fluid** acts
- as a lubricant, ensuring minimal friction during breathing.



- **Pleural effusion**
- **pleuritis**
- with too much fluid.

Lungs.



Right lung
Left lung
Anterior border
Apex
Hilum
Bronchus
Pulmonary artery
Pulmonary vein
Posterior border

structures the root = hilum of the lung

1. • **Bronchus:** contains air
2. • **Pulmonary artery**: carries **Un** oxygenated blood
3. • **2 pulmon**
   carries oxyg